

California's Interconnection Queue System and the Renewable Energy Transition:

Challenges, Reforms, and the Path Forward

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Abstract

The generator interconnection process (IP) has become a critical bottleneck in California's transition to a decarbonized energy system. As more renewable energy projects are proposed to meet state climate and reliability goals, the California Independent System Operator (CAISO) has faced mounting pressure to process an unprecedented number of interconnection requests. Despite serious structural and procedural reforms, the interconnection queue still lacks the efficiency and effectiveness necessary to support our evolving contemporary energy landscape. This undermines California's ability to integrate new clean energy resources, meet its climate targets, and maintain a predictable investment environment for developers.

This thesis evaluates the performance of the IP in California, focusing on the factors that contribute to the challenges it faces today. Drawing on public queue data, transmission planning documents, stakeholder comment letters, and semi-structured interviews with developers and regulators, the study uses both quantitative and qualitative methods to assess where and why interconnection bottlenecks persist. In addition to evaluating the interconnection process itself, the study explores how misalignment between state agencies contributes to the problem. The findings support a set of recommendations to improve coordination, reduce bottlenecks, and bring the process into closer alignment with the state's clean energy targets.

Table of Contents

Chapter 1: Introduction and Background.....	8
1.1 – Institutional and Process Background.....	9
1.2 – The California Energy Commission	10
1.3 – The California Public Utilities Commission.....	11
1.4 – The California Independent Systems Operator.....	12
1.5 – The Interconnection Process	13
1.6 – Organizational Cooperation	16
1.7 – Literature Review and Analytical Context.....	18
1.7.1 – Surging Queue Volumes and System Congestion.....	18
1.7.2 – Delays, Attrition, and Uncertainty	20
1.7.3 – Infrastructure Constraints and Planning Failures.....	20
1.7.4 – Financial Barriers and Developer Access	21
Chapter 2: Interconnection Queue and IPE Analysis.....	23
2.1 – Data Sources	23
2.1.1 – Analytical Approach.....	25
2.2 – Interconnection Request Trends and Cluster Volume Growth.....	27
2.2.1 – Analysis of the Withdrawn Projects Queue	33
2.3 – Delays	35
2.4 – Future Need and Current Decarbonization Trajectory	38

2.5 – IPE 2021 Track 1: Final Proposal Posted 4/21/22	42
2.5.1 – IPE 2021 Track 2: Final Proposal Posted 9/13/2022	43
2.6 – IPE 2023 Track 1: Final proposal posted 04/13/2023.....	49
2.6.1 – IPE 2023 Track 2: Final proposal posted 03/29/2024.....	50
2.6.2 – IPE 2023 Track 3: Final proposal posted 02/25/2025.....	60
2.7 – Non-CAISO Entity Policy Initiatives	64
Chapter 3: Interview Analysis.....	67
3.1 – Stakeholder Types	67
3.2 – Common Sentiments and Themes	73
3.2.1 – Stakeholder Perspective Comparisons.....	80
Chapter 4: Recommendations and Conclusion.....	82
4.1 – Expanded Stakeholder Data Exchange Platforms.....	82
4.2 – Increased Intra-Agency Integration of Regulatory Planning Initiatives	83
4.3 – Expanded Avenues for TPD Bidding	84
4.4 – Conclusion	85
References.....	89

List of Figures

Figure 1. Visual guide of CAISO’s interconnection process	15
Figure 2. Roles of key organizations in the interconnection process.....	17
Figure 3. Annual interconnection requests received by ISOs in the United States.....	19
Figure 4. Numer of active projects in the interconnection queue by cluster study. Note: Because Cluster 15 has not yet concluded, interconnection requests are shown instead.....	28
Figure 5. Withdrawn, active, and completed projects in the CAISO interconnection queue by cluster study.	29
Figure 6. Ratio of withdrawn projects to active and completed projects in the CAISO interconnection queue by cluster study.	29
Figure 7. Average time in years before project withdrawal by cluster study.	30
Figure 8. Average time in years between initial and current COD for active projects by cluster.	31
Figure 9. Number of projects by deliverability classification per cluster.	32
Figure 10. Project withdrawals by date since the start of the 2021 IPE process.	34
Figure 11. Delayed RPS-eligible vs non RPS-eligible resources (28.4 GW total.	36
Figure 12. Visualization of Table 2	39
Figure 13. Visualization of Table 3	Error! Bookmark not defined.
Figure 14. Visualization of withdrawals occuring after September 2023 final decision announcement	46
Figure 15. Visualization of Table 4	47
Figure 16. Visualization of Table 5	48
Figure 17. Visualization of Table 6	49
Figure 18. CAISO published map depicting the new interconnection zones	52

Figure 19. Total MW of withdrawn vs non-withdrawn Cluster 15 projects by POI; pie chart shows TPD headroom at POIs for each category	55
Figure 20. CAISO heatmap as published onwabesite.....	58
Figure 21. Visualization of Table 8	58
Figure 22. Close-up of project withdrawals between 2024 and 2025; small clusters emerge in late 2024, loosely coinciding with the start of IPE 2023 Track 3	62
Figure 23. Visualization of Table 9	62

List of Tables

Table 1. Types of delays impacting planned generation projects.....	37
Table 2. IQ Active Queue Interconnection Capacity in MW by Generation Type	39
Table 3. Cluster 15 Interconnection Requests in MW by Generation Type.....	40
Table 4. Stakeholder sentiment for minimum PPA Requirements	47
Table 5. Higher Fees, Deposits, & Other Criteria for Submitting an IR.....	48
Table 6. Pausing Cluster 15 and postponing Cluster 14 Phase 2	49
Table 7. New CAISO proposed TPD Allocation Scoring Criteria.....	53
Table 8. Sentiment for new TPD Allocation structure	58
Table 9. Developer sentiment regarding changes to parked project standards	62
Table 10. Developer sentiment regarding possible improvements to IC process	69
Table 11. Policy Regulator sentiment regarding possible improvements to IC process	71
Table 12. LSE sentiment regarding possible improvements to IC process.....	73
Table 13. Interview Summary: Stakeholder Sentiments.....	77

Commonly Used Acronyms

CAISO – California Independent Systems Operator

CPUC – California Public Utilities Commission

CEC – California Energy Commission

FERC – Federal Energy Regulatory Commission

FCDS – Full Capacity Deliverability Status

GIA – Generator Interconnection Agreement

IPE – Interconnection Process Enhancement

IQ – Interconnection Queue

IR – Interconnection Request

LSE – Load Serving Entity

PPA – Power Purchase Agreement

RA – Resource Adequacy

RPS – Renewable Portfolio Standard

RNU – Reliability Network Upgrade

TPD – Transmission Plan Deliverability

Chapter 1: Introduction and Background

As our energy needs have evolved throughout the start of the 21st century, nations around the world have recognized the need to transition to more sustainable forms of generation. In California, legislation such as Senate Bill 32 and Senate Bill 100 have established clear renewable energy procurement goals and greenhouse gas emission limits, shaping the behavior of developers, regulators, and other stakeholders. Central to this transition is the integration of new renewable energy resources into the main electrical grid, a process known as generator interconnection. Interconnection refers to the regulatory and technical steps required for a power generation facility to connect to the transmission system so that the electricity it produces can be delivered to market. It is a complex, multi-step process that involves engineering studies, legal contracts, economic modeling, and the coordination of multiple agencies and private entities.

However, as the number of renewable projects grows, the system that handles these interconnections has started to crack under pressure. CAISO, which oversees most of the state's grid operations, has been flooded with new applications, leading to delays, backlogs, and mounting frustration for developers (CAISO, 2024). These slowdowns make it harder for California to stay on track with its clean energy goals. Reforms like CAISO's Interconnection Process Enhancements (IPE) and FERC Order 2023 have tried to streamline the process, but the true extent of their impact is yet to be observed.

What has become more apparent in recent years however, is that the bottlenecks aren't just about red tape or outdated procedures. As agencies and market actors are asked to take on expanded roles, the absence of consistent data transparency and clearly defined accountability has led to

friction. Fixing California's interconnection issues will require more than just technical tweaks. Institutions need clearer roles and better coordination to reflect the realities on the ground.

This thesis examines the IP as it exists in California, scrutinizing institutional gaps, the procedural chokepoints, and the mismatches in planning that lead to project attrition and delay. Drawing on IQ data, regulatory documents, stakeholder comment letters, and interviews with developers and agency staff, it aims to evaluate the efficacy of recent reforms and offer feasible recommendations for improving interconnection outcomes in future cluster cycles.

1.1 – Institutional and Process Background

In today's modern grid system, there exist a multitude of organizations and stakeholders that participate in the IP. The IP refers to the best practices and procedures that new generation resources such as gas plants or wind turbines must follow to be connected to the greater electrical grid and thus be able to deliver the energy they generate to their customers. While at first glance it may seem a straightforward endeavor, it is a heavily regulated industry with many requirements that must be met before developers are able to open their projects and begin generating electricity to sell.

These are a direct result of the unique factors present in the energy sector. The high cost of interconnection infrastructure, the natural monopoly status of utility companies, the transmission capacity limits of our existing infrastructure, the requirements needed for a new generation facility to be studied for possible interconnection, California's need to meet its legislatively mandated renewable energy standards, the list of unique factors that must be considered goes on and on. Adding to these complications is the fact that there exist multiple regulatory agencies

participating directly and indirectly in the process. Each one has specific areas of expertise and delegated responsibilities it must consider, with even more complexities arising from the interplay between these agencies. To best analyze the current situation regarding generator interconnection, it is crucial to also understand the authority these various organizations wield, and how each organization has agreed to work together to achieve their long-term objectives.

1.2 – The California Energy Commission

While playing a lesser role in the IP, the CEC still maintains a vital role in the broader context of the California energy landscape which has downstream impacts on the determinations of the CPUC and CAISO. The most relevant of these roles relating to interconnection lies in their forecasting of California's future energy demand through its Integrated Energy Policy Report, a biennial publication that provides detailed projections for electricity, natural gas, and transportation energy needs (CEC, 2024). These forecasts act as inputs for a variety of other policies and programs such as the Resource Adequacy program, the Renewable Portfolio Standard, and the Transmission Planning Process.

Additionally, the CEC's scenario planning, which explores various future energy demand via predictive modeling helps the CPUC and CAISO anticipate potential challenges and opportunities in the IP. For instance, if the CEC identifies a high-growth scenario for renewable energy deployment, the CPUC may prioritize interconnection for projects that align with this scenario. CAISO meanwhile may accelerate transmission upgrades to accommodate the increased flow of renewable energy. Conversely, if the CEC's forecasts indicate slower growth in

certain sectors, the CPUC and CAISO may adjust their interconnection priorities accordingly to ensure that resources are allocated efficiently and cost-effectively.

1.3 – The California Public Utilities Commission

Organizations that participate in the energy sector of California are subject to oversight and regulation by the CPUC, which as their name implies is responsible for regulation of Californias public utilities. This oversight comes in a variety of ways, such as approving or barring changes in ratepayer prices and ensuring that load serving entities (LSEs) are procuring adequate energy from renewable sources to be in compliance with state renewable energy standards.

While they are not able to exert a direct influence on the decisions that CAISO make regarding their transmission planning process and policy modifications, many of the decisions that the CPUC make have far reaching downstream effects that play a significant role in the IP. The interconnection resource plan (IRP) refers to the CPUCs planning proceedings ensuring that California is able to meet its energy demands safely, reliably, cost-effectively, and in a way that is consistent with our green energy goals.

Briefly summarized, each regular IRP publication contains:

1. Preferred System Plan (PSP) Portfolios

A statewide resource portfolio that meets forecasted load, reliability, and GHG goals, guiding what resources (solar, wind, battery, geothermal, etc.) should be built.

2. LSE-Specific Procurement Requirements

Based on the PSP, the CPUC can mandate specific quantities and types of resources each LSE must procure. These can include long-duration storage, firm zero-carbon capacity like geothermal, or generic clean MW targets.

3. Transmission Planning Inputs

The CPUC compiles and provides approved portfolios to the CAISO's Transmission Planning Process, influencing where transmission upgrades should be made to unlock future generation and ensure that long term planning goals are being met.

Each step of the IP is influenced in some ways by the above inputs. Before interconnection requests for the latest cluster cycle are even solicited, CAISO must determine where new transmission upgrade projects will be triggered via the input provided to them for the TPP. When these transmission projects are completed, some of the deliverability may be reserved to meet portfolio standards. When and what resources are selected for procurement can influence which types of projects are more able to secure a power purchase agreement (PPA), which acts as an important input to the scoring process for determining which generation projects will be studied for interconnection and allocated transmission deliverability.

1.4 – The California Independent Systems Operator

CAISO is the most prominent of these organizations when discussing the IP, playing a central role in its planning and implementation. Their primary responsibilities consist of planning, soliciting, and integrating transmission infrastructure and new generation resources into

California's energy grid. Unlike the CEC and the CPUC, CAISO is not a California governmental entity but rather a nonprofit public benefit corporation authority that is neither fully public nor private. Its authority stems from its role as the operator of the state's bulk electric system and wholesale electricity market, granting it the responsibility to ensure that new generation projects are interconnected in a way that maintain both grid reliability as well as alignment with California's energy and climate goals. It is under the official jurisdiction of the Federal Energy Regulatory Commission (FERC), and any changes that CAISO makes to its official tariffs require FERC's authorization.

1.5 – The Interconnection Process

CAISO's IP is structured around cluster studies, which allow multiple project proposals to be evaluated simultaneously. Each cluster cycle, developers submit interconnection requests during a designated application window. All qualifying projects are grouped into a single "cluster" and studied together to assess their collective impact on the transmission system. This approach has many benefits over the previous first come, first serve structure that most ISO's utilized in the past, allowing for assessments of systems with far greater levels of complexity.

1. Interconnection Request (IR) Submission

Potential developers submit a formal request to CAISO, consisting of technical specifications of the project, financial details, necessary fees, and more.

2. Study Process Selection

CAISO performs an analysis of the requests received, assigning a score to each potential project which will determine if the submitted requests will proceed to the study phase.

3. *Scoping Meeting*

After a determination for projects eligible for study has been made, CAISO meets with the prospective developers to review their submission.

4. *Interconnection Studies*

Once both parties have met and an understanding has been reached about the specifications of the proposed project, a variety of engineering related studies are conducted to determine the feasibility of interconnection and the nature of infrastructure upgrades and construction that would be needed to accommodate it.

5. *Study Result Analysis*

An analysis of the study results is conducted, providing an opportunity for the interconnection customer to clear up any discrepancies or clarify any findings.

6. *Interconnection Agreement (IA) Negotiation*

Once the results of the study have been fully considered and approval provided by CAISO, an IA is drafted and contracts agreed upon detailing construction timeframes, generator specific interconnection procedures, and other project specific variables.

7. *Entrance to Active Queue*

The project finally enters the “Active” IQ, with a predicted commercial operation date and will remain classified as such until the project either sees completion or withdraws from the queue.

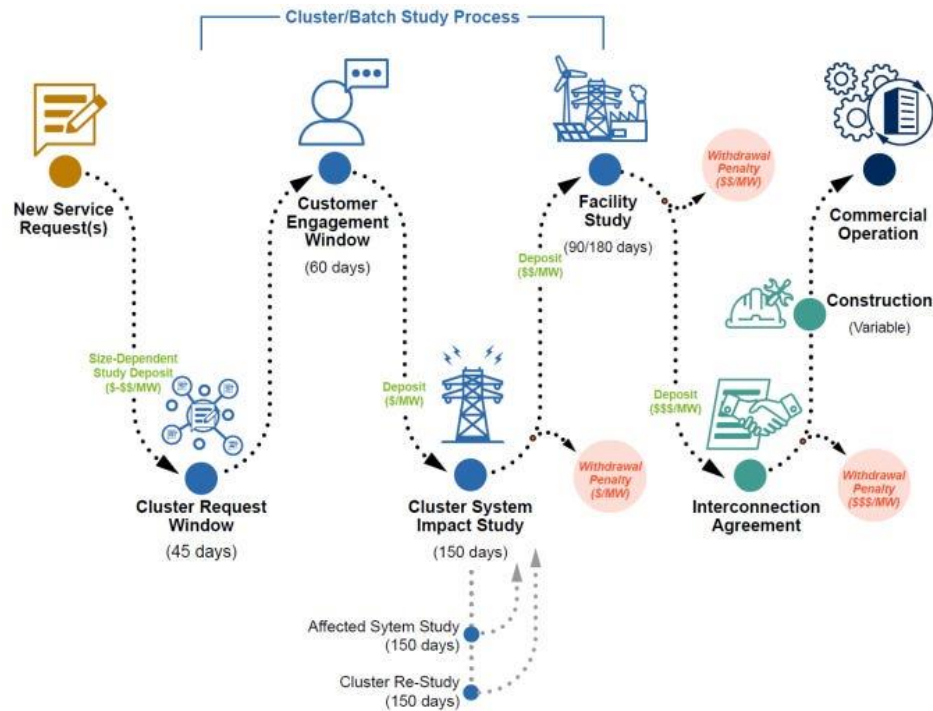


Figure 1. Visual guide of CAISO's interconnection process

In addition to standard technical evaluation, projects seeking Full Capacity Deliverability Status (FCDS) or Partial Capacity Deliverability Status (PCDS) must apply for Transmission Plan Deliverability (TPD) allocations. TPD refers to the amount of capacity available at substations during peak congestion periods and determines whether a project's generation can be classified as resource adequacy capacity, a key requirement for securing long-term contracts.

TPD allocations are awarded based on a scoring system that evaluates project readiness across several factors, including permitting status, engineering design, and material acquisition. Projects that do not receive deliverability are reclassified as energy-only, meaning they can only sell electricity during uncongested hours and may face lower financial viability. Developers may reapply for TPD in future allocation cycles if eligibility criteria continue to be met.

1.6 – Organizational Cooperation

To summarize, the CEC, CPUC, and CAISO collaborate as independent organizations with many interdependent functions, working toward the goal of interconnecting generation projects to the grid while also ensuring that the RPS standards set by the state are being met. The CEC's Integrated Energy Policy Report (IEPR) provides critical forecasts and scenario planning that inform the CPUC's Busbar Mapping efforts, which in turn guide CAISO's transmission planning and interconnection studies. Decisions made by the CEC, such as identifying high-growth renewable energy scenarios, directly influence the CPUC's prioritization of interconnection projects and CAISO's allocation of resources for grid upgrades. Similarly, CAISO's interconnection studies and cost allocation decisions, governed by its Generator Interconnection Procedures (GIP), feed back into the CPUC's oversight of renewable portfolio standard (RPS) compliance and the CEC's long-term energy planning.



Figure 2. Roles of key organizations in the interconnection process

While this iterative and interdependent process ensures that each organization's actions are aligned with California's renewable energy goals, it also means that delays or inefficiencies in one agency's operations can ripple through the entire system. These ripples have great potential in impacting the deployment of renewable energy and the state's ability to meet its decarbonization targets beyond the immediate and obvious implications. Therefore, when potential solutions or optimizations to the overall system are proposed, they must factor in which parts of the system they will impact the most as well as what sort of downstream effects may emerge because of their implementation. While this does not lend itself to a straightforward and simple solution, it does mean that there exist various avenues through which changes can be implemented.

1.7 – Literature Review and Analytical Context

As the pace of renewable energy deployment accelerates, the IP has come under increasing scrutiny. Originally designed for a more centralized and predictable energy landscape, current interconnection procedures are increasingly mismatched with the scale, speed, and complexity of modern clean energy development. This chapter reviews the existing literature on the evolution of the IP, the challenges that have emerged, and the limitations of recent reform efforts. It also identifies key areas of contention and gaps that remain underexplored, particularly as they relate to California’s regulatory and transmission planning landscape.

Four key themes recur throughout the literature. First, researchers emphasize the unprecedented volume of projects entering IQs, driven by policy mandates and declining technology costs, and the resulting strain on transmission planning and review processes. Second, there is growing concern about the increasing timelines and rising attrition rates within queues, especially for large-scale or first-time developers. Third, literature points to structural and procedural barriers such as fragmented grid oversight and uncoordinated upgrade planning as major contributors to delay and uncertainty. Finally, scholars and stakeholders alike have raised concerns about equity and access, noting that escalating financial requirements risk excluding smaller developers and distorting market participation.

1.7.1 – Surging Queue Volumes and System Congestion

Over the past decade, the volume of proposed renewable energy projects has increased dramatically across U.S. transmission networks, placing significant strain on interconnection

procedures administered by independent system operators. The primary drivers of this surge include the declining cost of wind, solar, and storage technologies, alongside aggressive state and federal decarbonization targets (Jayabal, 2024). As can be seen in the figure below, by the end of 2023, total capacity in U.S. IQs exceeded 2,600 GW, more than double the volume seen just five years earlier. CAISO alone accounted for approximately 523 additional GW of capacity, reflecting the scale of developer interest and the mismatch between generation ambition and transmission capability (Howland, 2024).

Notably, over 95% of queued projects now involve solar, wind, or energy storage resources (Mills, 2024). These technologies often require novel grid integration strategies, given their intermittency, geographic distribution, and preference for siting in remote high-resource areas. The shift from centralized fossil-fuel generation to dispersed renewables has outpaced the grid's physical and institutional ability to accommodate change, producing what many scholars characterize as a structural bottleneck rather than a temporary backlog (CAISO, 2022).

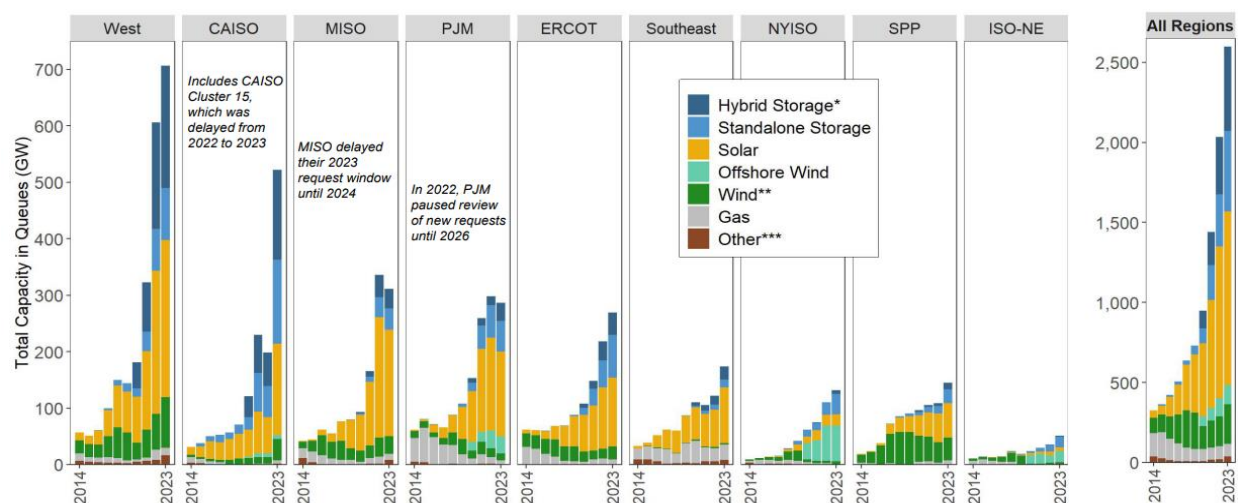


Figure 3. Annual interconnection requests received by ISOs in the United States

1.7.2 – Delays, Attrition, and Uncertainty

As queues grow, interconnection timelines have lengthened. Median project durations from initial request to commercial operation rose from approximately four years during 2018–2022 to over five years by 2023 (DeLosa, 2024). These extended timelines introduce significant financial uncertainty for developers, who must often wait years before knowing the cost and feasibility of interconnection upgrades. As a result, project attrition remains high and in some ISOs a majority of proposed projects are withdrawn before reaching completion.

Empirical studies show a strong correlation between interconnection delay and project cancellation, especially in regions with high renewable penetration and complex permitting environments (California Public Utilities Commission, 2024; Energy Markets & Policy, 2024). In PJM, for example, 90% of surveyed developers cited interconnection challenges as the leading barrier to commercial operation (Driscoll, 2022). Delays are especially acute for hybrid or large-scale renewable projects, which typically require both transmission upgrades and extensive environmental review (Wilson, 2024).

1.7.3 – Infrastructure Constraints and Planning Failures

The scale of proposed generation far exceeds the capacity of current transmission planning mechanisms, many of which were not designed to accommodate the sudden influx of distributed renewable development that has taken place in the past decade. Literature highlights a recurring problem of reactive, outdated transmission upgrade processes that lag behind interconnection demand (Farmer, 2024). Without coordinated regional or inter-ISO planning, projects often

trigger localized upgrades such as new substations or feeder lines that are technically feasible but economically inefficient when considered in isolation.

For example, in California, despite efforts to coordinate its Transmission Planning Process (TPP) with its Interconnection Study Process, CAISO continues to assess interconnection and transmission upgrades largely in parallel. As a result, opportunities for efficient and optimal solutions are often missed. Even recent proposals under the 2023 IPE Track 2 reforms have been criticized by developers for reinforcing a utility-centric model that favors deliverability to substations that already possess additional transmission capacity, potentially sidelining distributed or community-scale projects (Howland, 2024). This lack of proactive planning is exacerbated by fragmented jurisdiction across federal, state, and ISO boundaries. While FERC has called for greater long-term planning under Order 1000 and related reforms, implementation has been inconsistent, in part due to the lack of enforceable interregional coordination mechanisms and varying levels of state participation. (FERC, 2024; CAISO, 2023).

1.7.4 – Financial Barriers and Developer Access

Multiple sources identify initial financial requirements as a key source of inequity in the IP. Network upgrade costs, which are often unpredictable and imposed late in the IP, represent a particularly high burden for small or first-time developers (Orrick, 2024). These upfront risks discourage market entry and contribute to growing consolidation in renewable development. Recent reforms, including FERC Order 2023, attempt to increase cost transparency and introduce enforceable study timelines, but their long-term impact remains uncertain.

The intersection of rising infrastructure costs, policy-driven urgency, and regulatory lag has created an environment characterized by long timelines, high attrition, and reduced predictability (Rand, 2024). Without more comprehensive planning and governance reforms, even well-capitalized projects may fail to reach commercial operation in time to meet state and federal climate goals (Lala, 2023; Rand, 2024). Taken together, the literature highlights an interconnection system struggling under both procedural and structural weight. While recent reforms offer some relief, their long-term effectiveness is still too early to assess.

Chapter 2: Interconnection Queue and IPE Analysis

California's interconnection process has come under intense scrutiny in recent years, as the pace and scale of new generation proposals outstrips the state's ability to process and integrate them efficiently. This chapter examines the performance of the IP through both empirical and qualitative lenses, focusing on trends in project attrition, delays, and deliverability challenges across Clusters 13 through 15. In parallel, it assesses the policy rationale and early impact of recent reforms introduced under CAISO's IPE initiative. Drawing on queue data, stakeholder filings, interviews, and planning documents, this chapter aims to evaluate whether California's interconnection system is adapting effectively to meet its clean energy goals.

2.1 – Data Sources

This thesis draws on four primary categories of data, each serving to act as a foundation from which observations and connections can be made regarding project-level outcomes, regulatory intentions, and stakeholder experience.

1. Interconnection Queue Dataset

The public interconnection queue, maintained by CAISO, provides a record of all generation projects requesting grid access. Fields include application dates, capacity, withdrawal status, commercial operation timelines, and deliverability classifications. This dataset forms the basis for quantitative trend analysis in Clusters 13-15, enabling measurement of queue volume,

attrition rates, delays, and shifts in deliverability strategy. While it does not capture the full reasoning behind individual project outcomes, it allows for identification of broader trends in project attrition, delays, and shifts in deliverability strategy across Cluster cycles.

2. Stakeholder Comment Letters

Formal comment letters submitted during CAISO's IPE initiatives serve as a key source of stakeholder perspectives. These submissions contain the opinions and sentiments of developers, LSEs, utilities, and advocacy groups, and can offer direct insight into stakeholder perspectives. Because they are tied to specific regulatory proceedings, they reveal how policy proposals are interpreted and contested in real time. To evaluate stakeholder sentiment on IPE Track 1 proposals, this study analyzed public comment letters submitted during the relevant CAISO stakeholder process. Each comment was reviewed and manually coded based on the general position expressed toward the proposed delay of Cluster 14 study activities. Organizations were grouped into broad categories based on their role in the interconnection process, and their sentiments were sorted into five qualitative classifications: Agree, Somewhat Agree, Neutral, Somewhat Oppose, and Oppose. These categories were chosen to capture both clear support or dissent as well as more qualified or mixed responses, which often included conditional support or specific caveats. This classification scheme enables comparison across stakeholder types while preserving the nuance of individual responses.

3. Semi-Structured Interviews

To supplement documentary sources, interviews were conducted with a targeted sample of developers, regulators, and LSE representatives. Given the complexity of the interconnection process and the limitations of publicly available data, interviews were chosen to capture the lived experiences, institutional dynamics, and informal practices that shape how interconnection policy plays out in practice. While regulatory documents and stakeholder comments provide insight into formal positions and procedural reforms, they often lack the nuance and context needed to understand how policies are interpreted and operationalized by those directly involved. Interviews allowed for exploration of institutional frictions, interpretive differences, and emerging concerns that may not be reflected in official filings. This qualitative input was especially valuable in evaluating areas where institutional misalignment and policy implementation challenges contribute to project delays and stakeholder dissatisfaction.

4. Planning and Regulatory Documents

These documents serve as the connective tissue between regulatory goals and the actual design of the interconnection process. While they don't always reflect how policy plays out on the ground, they provide essential context for understanding why certain decisions were made, how institutional responsibilities are distributed, and where gaps or misalignments might emerge. This category of material was especially useful for grounding stakeholder interviews and queue analysis in the broader regulatory framework they operate within.

2.1.1 – Analytical Approach

This thesis applies a mixed-methods approach to assess the performance of California's interconnection process. Quantitative analysis centers on four key indicators drawn from CAISO queue data:

1. Queue Volume: The number of projects submitted and remaining in the queue, reflecting administrative capacity and developer interest.
2. Withdrawal Rates: The proportion and timing of project withdrawals, serving as an indicator of screening efficacy and developer confidence.
3. Project Delays: The average time between a project's initial and current commercial operation date, highlighting bottlenecks and uncertainty.
4. Deliverability Status: The proportion of projects requesting Full or Partial Capacity Deliverability Status (FCDS, PCDS), which affects their access to long-term contracts and financial viability.

These indicators were chosen to evaluate procedural throughput and developer behavior in response to institutional reforms. While detailed queue processing timelines were not available, proxies such as shifts in commercial operation date and withdrawal timing were used to infer systemic delay.

Qualitative insights were drawn from two main sources, formal stakeholder comment letters submitted to CAISO during the IPE reform process and semi-structured interviews with developers, LSEs, and regulators. A thematic coding approach was used to identify points of convergence and friction, with particular attention to transparency, regulatory stability, and project viability, as well as recurring concerns around deliverability and institutional

coordination. These qualitative inputs serve to contextualize quantitative trends and assess the perceived effectiveness of recent policy changes.

Two additional strategies were employed: (1) policy evaluation using a limited comparative analysis of Clusters 13–15 to observe early indicators of reform impact, and (2) geospatial visualization, overlaying queued project locations with transmission infrastructure maps to identify regional bottlenecks.

2.2 – Interconnection Request Trends and Cluster Volume Growth

1. Number of projects in IQ

The number of projects entering CAISO’s IQ has increased steadily with each new cluster cycle, with a particularly sharp rise observed between Clusters 13 and 14. This trend has placed a growing burden on CAISO’s resources, as each project requires detailed technical evaluation and modeling. The scale of incoming requests also contributes to greater grid complexity, as more variables must be considered from both a reliability and system operations standpoint.

Figure 1 shows a steep rise in active projects between Clusters 13 and 14, with Cluster 15 maintaining similarly high volume. Although withdrawals are likely to reduce the final number of active Cluster 15 projects, early indicators suggest the queue will remain at or near the scale of Cluster 14. Given California’s ambitious clean energy targets, future cluster volumes are unlikely to decline, underscoring the importance of a more scalable and efficient IP.

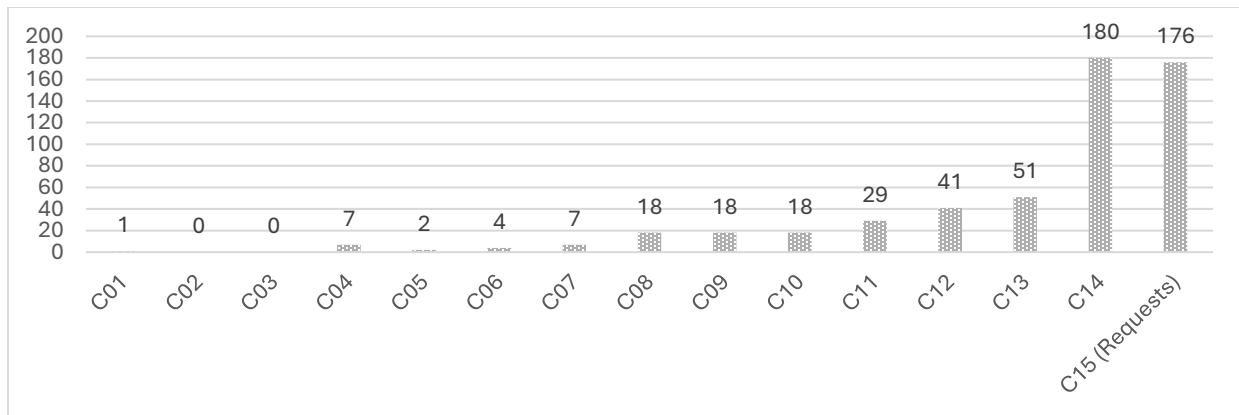


Figure 4. Numer of active projects in the interconnection queue by cluster study. Note: Because Cluster 15 has not yet concluded, interconnection requests are shown instead.

Source: CAISO IQ, retrieved May 1, 2025

2. Withdrawn Projects

In addition to tracking active projects, the number of withdrawn interconnection requests provides important insight into the overall health and efficiency of the IP. While the total number of projects in Cluster 14 is significantly higher than in previous clusters, the volume of withdrawals has not increased proportionally. This suggests that CAISO’s screening process may be improving, selecting projects with a higher likelihood of advancing to later study phases.

As seen in Figure 2, despite the surge in Cluster 14 projects, withdrawal rates remained relatively stable. This may suggest improved front end filtering or more strategic developer submissions. While projects in the active queue from each subsequent cluster will naturally be smaller due to the increasing demand and reclassification of projects that have been completed from “active” to “complete”, the jump we see in Cluster 14 is unusually large. Given that no projects from Clusters 12 or 13 have reached completion, the sharp increase in Cluster 14 actives suggests a backlog, rather than improved throughput.

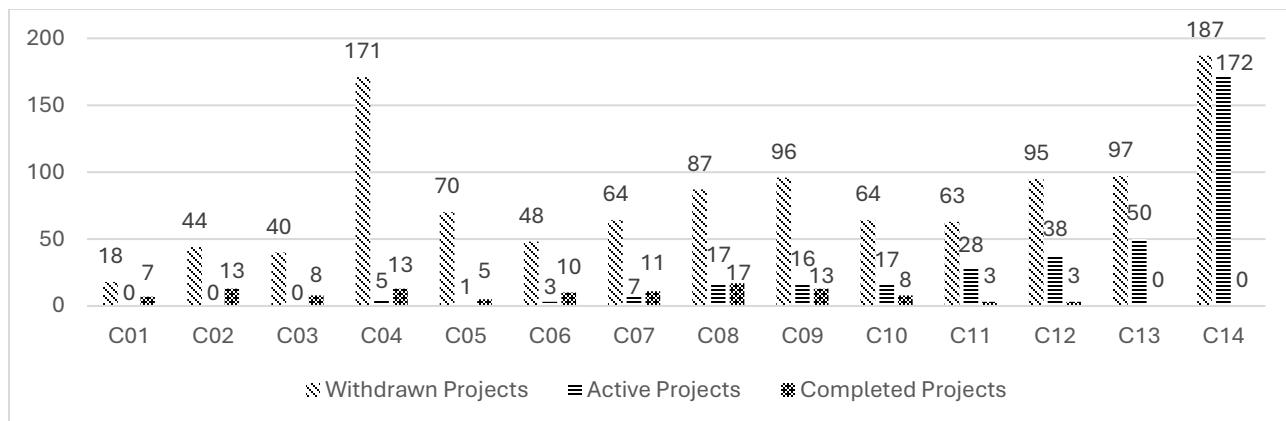


Figure 5. Withdrawn, active, and completed projects in the CAISO interconnection queue by cluster study.

Source: CAISO IQ, retrieved May 1, 2025

The ratio of withdrawn projects to those that reached active or completed status provides a secondary indicator of queue quality. Cluster 14 continues a trend of lower withdrawal ratios compared to earlier cycles, though the fact that it is the most recent cluster study most likely plays some part in that as well. However, it is a significant enough decrease when compared to previous studies which may suggest improved project screening or more cautious developer behavior.

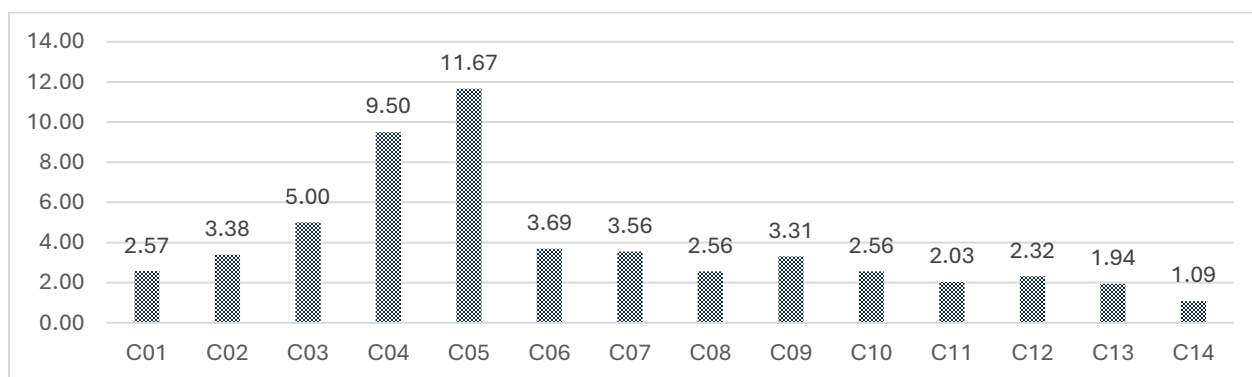


Figure 6. Ratio of withdrawn projects to active and completed projects in the CAISO interconnection queue by cluster study.

Source: CAISO IQ, retrieved May 1, 2025

A second metric of system performance is the average duration that withdrawn projects remain in the queue. Projects in the queue act as inputs for CAISO’s transmission planning models and incorporated into CPUC-led programs like Resource Adequacy and the Renewable Portfolio Standard. If these projects ultimately fail, prolonged inclusion in the queue distorts planning forecasts and misallocates modeling resources. Earlier withdrawal reduces systemic noise and improves long-term forecasting accuracy.

Cluster 14 withdrawals remained in the queue longer, on average, than withdrawals from any previous cluster, as shown in Figure 4. Because Cluster 14 is ongoing, additional withdrawals may further increase this average. Even so, the current duration points to a lag in early-stage project screening, a potential area for process improvement.

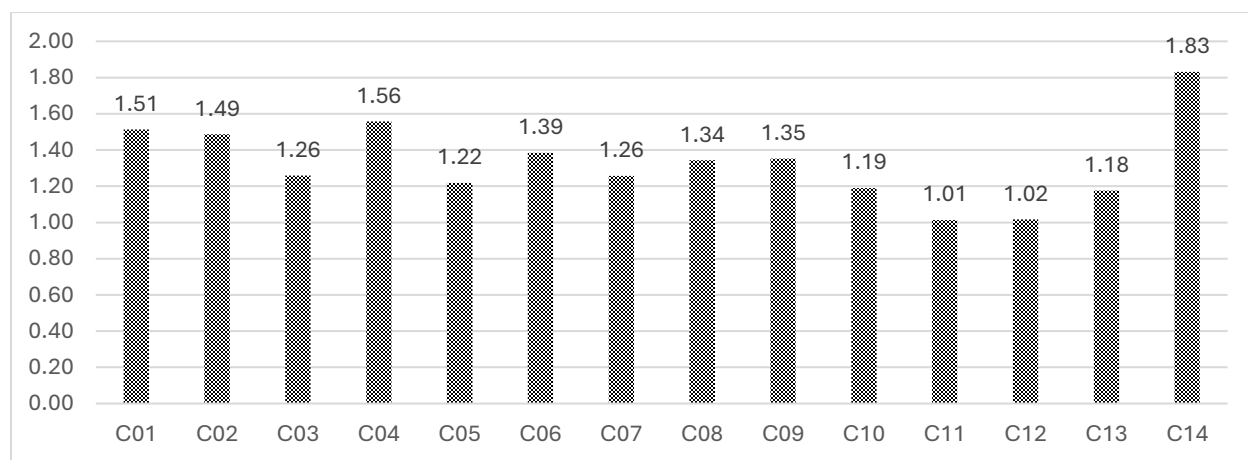


Figure 7. Average time in years before project withdrawal by cluster study.

Source: CAISO IQ, retrieved May 1, 2025

3. Magnitude of delays and number of “parked” projects

Alongside rising queue volumes, average project delays have also grown sharply between Clusters 14 and 15. In Cluster 14, the delta between Initial COD and Current COD listed in the

CAISO IQ is greater than in prior clusters, suggesting project schedules are slipping earlier in the development cycle. While there is no obvious cause for this, it is likely due to factors such as supply chain disruptions, delayed transmission upgrades, or shifting economic conditions. Regardless of cause, the trend suggests that developers are either encountering obstacles earlier in the development timeline or overestimating how quickly they can bring projects online. In congested zones, project timelines are interdependent. Delays for one generator can trigger cascading effects across other nearby projects, especially when transmission capacity is limited. CAISO rules also allow projects to retain queue positions under certain conditions, effectively 'parking' their interconnection rights. While this avoids immediate penalties, it contributes to queue stagnation by enabling developers to delay without progressing.

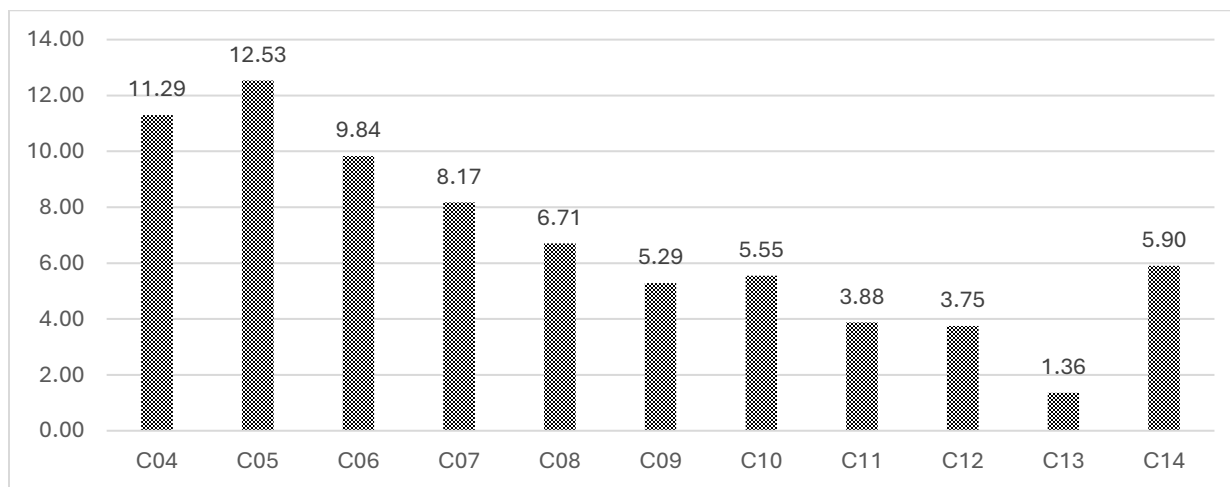


Figure 8. Average time in years between initial and current COD for active projects by cluster.

Source: CAISO IQ, retrieved May 1, 2025

4. Deliverability Status

A core consideration for projects entering the IQ is whether they will seek transmission deliverability status, which determines their ability to supply energy during peak load periods.

Deliverability status, which is classified as either *Full Capacity Deliverability Status* (FCDS) or *Partial Capacity Deliverability Status* (PCDS), is not just a technical designation. It directly affects whether a project can qualify as resource adequacy (RA) capacity and secure long-term PPAs, making it a central financial and operational concern for developers.

As shown in Figure 4, Cluster 14 saw a sharp increase in the number of projects applying for FCDS, even as energy-only and PCDS requests remained relatively flat. As observed, the number of energy-only projects **declined**, despite the overall queue growing substantially in that cycle. This suggests a growing preference among developers for projects that can participate in capacity markets and offer firm deliverability guarantees; possibly in response to regulatory signals or market incentives.

Because deliverability studies require additional modeling, deposits, and coordination, this shift increases the workload for CAISO and adds strain to the system. The rise in FCDS requests has become a focal point of recent interconnection reform efforts, particularly under the 2023 IPE, which aim to refine how deliverability is evaluated and awarded.

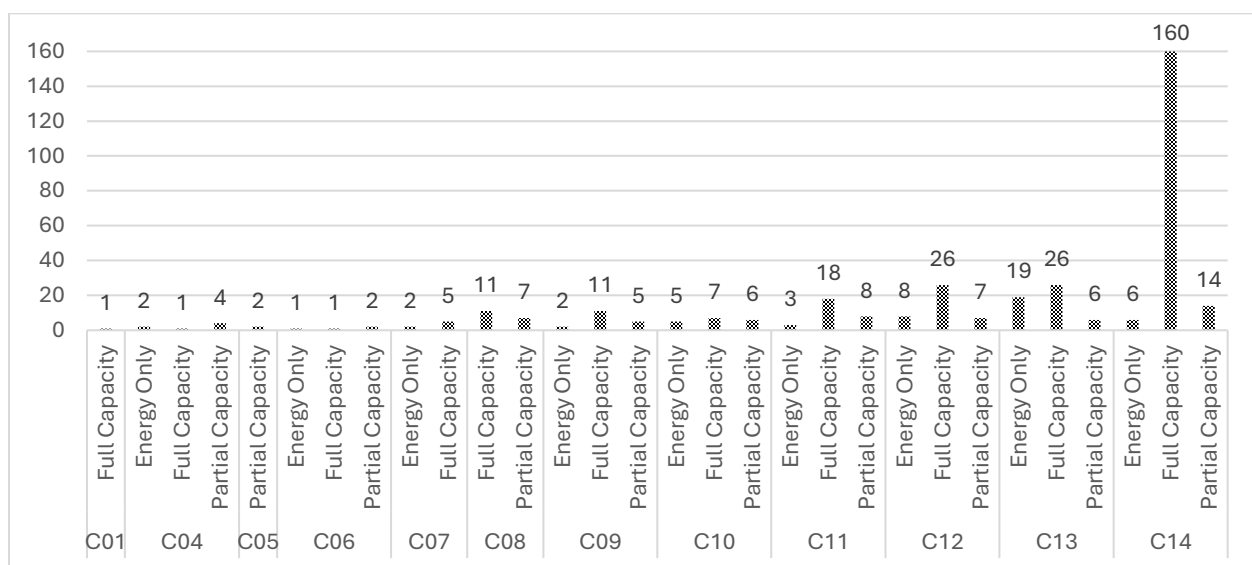


Figure 9. Number of projects by deliverability classification per cluster.

Source: CAISO IQ, retrieved May 1, 2025

While Cluster 14 concluded with the successful study and posting of results, it also highlighted persistent stress points in California’s interconnection system. High volumes of queued projects, rising withdrawal rates post-study, and uneven project advancement suggest continued inefficiencies. Two key performance indicators (reducing post-study withdrawals and increasing project completion) remain simple yet powerful metrics for evaluating the effectiveness of interconnection reform. However, identifying the mechanisms by which these outcomes should be achieved is far more complex.

CAISO’s IPE initiative represents a key tool for refining this process. Through regular, iterative updates to its tariff language, CAISO aims to address procedural bottlenecks and align the interconnection framework with California’s broader energy policy objectives. The 2021 and 2023 IPE initiatives, each divided into multiple “tracks,” introduced substantial procedural changes, with a focus on the details of how projects are screened, studied, and assigned deliverability.

In parallel, FERC Order 2023 imposed new nationwide standards for generator interconnection, requiring all transmission providers to adopt improvements related to study timelines, cost transparency, and project readiness criteria. CAISO’s Track 2 IPE reforms and FERC Order 2023 share overlapping goals but differ in scope and implementation.

2.2.1 – Analysis of the Withdrawn Projects Queue

Withdrawn projects offer a valuable window into queue dynamics, particularly when assessed over time. Although withdrawal is not an ideal outcome, signaling developer uncertainty, unmet

readiness criteria, or shifting market conditions, it can serve as a corrective mechanism if it occurs early in the process before study resources are spent.

One of CAISO's goals under both the 2021 and 2023 IPE has been to reduce queue congestion by disincentivizing speculative or non-viable projects. As a result, we would expect withdrawal behavior to become more predictable over time, with fewer late-stage exits and less erratic surges.

Figure 7 below visualizes the cumulative number of project withdrawals from the IQ starting in early 2022, prior to the implementation of Track 2 of the 2021 IPE. Periods of steep slope in the graph correspond to a large numbers of withdrawals in a short timeframe, while flatter segments suggest a slowdown in withdrawal activity. The smoothing out of withdrawal spikes in the most recent periods (particularly in late 2023 and early 2024) may reflect improvements in CAISO's initial project screening, as well as the impact of readiness and financial requirements introduced under the IPE.

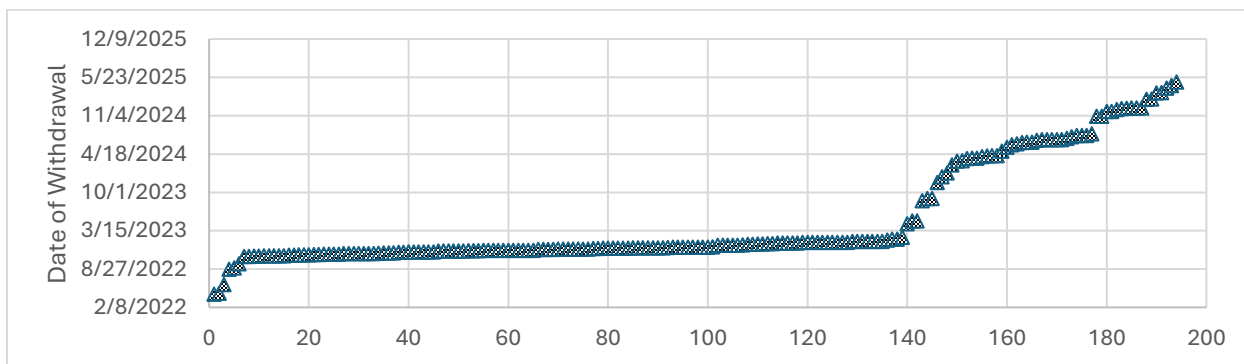


Figure 10. Project withdrawals by date since the start of the 2021 IPE process.

Source: CAISO IQ, retrieved May1, 2025

These observations are preliminary, and the data remains incomplete for Cluster 15 and beyond.

However, if sustained, the trend would suggest that reforms are beginning to reduce late-stage attrition and encourage earlier self-selection out of the process.

2.2.2 – Changing Cluster Study 15 Requests

While the withdrawn projects list offers insight into how IPE reforms are shaping developer behavior, there are limitations when analyzing Cluster 15. Unlike active queue withdrawals, which are archived with clear dates and metadata, Cluster 15 withdrawals, specifically those that occurred before projects formally entered the active queue, are not systematically recorded by CAISO. As a result, the only reliable data source for withdrawn Cluster 15 requests has come from prior snapshots of the queue collected during the course of this research.

This lack of timestamped withdrawals with informative metadata makes it difficult to attribute changes in developer behavior to specific policy milestones, such as the implementation of Track 2 IPE reforms. Nonetheless, trends within the Cluster 15 application window still offer clues. Because several major IPE changes were designed to take effect starting with Cluster 15, developer response in this cohort may provide a first look at the efficacy of these policy shifts. In particular, shifts in the composition of project types (e.g., fewer speculative proposals, more FCDS-seeking applications) may reflect increased selectivity and earlier self-screening by developers.

2.3 – Delays

Project delays remain one of the most persistent obstacles to efficient interconnection. Because CAISO’s IP is cyclical and iterative, delays in one project can produce cascading effects across multiple clusters. These can arise from a range of causes: material shortages, permitting complications, natural disasters, or stalled transmission infrastructure upgrades. Recent queue data show that the gap between initial proposed Commercial Operation Dates (CODs) and currently expected CODs has widened significantly in Cluster 14, with many projects already registering delays despite their relative recency. This may signal either increased external pressures on developers or continued over-optimism in initial COD estimates. In either case, delays introduce uncertainty not just for individual developers but for the broader system, particularly in zones where transmission constraints link multiple interdependent projects.

**Figure 15: RPS Eligible vs Non-RPS Eligible Resources (28.4 GW)
from 2024 SB 1174 Reporting**

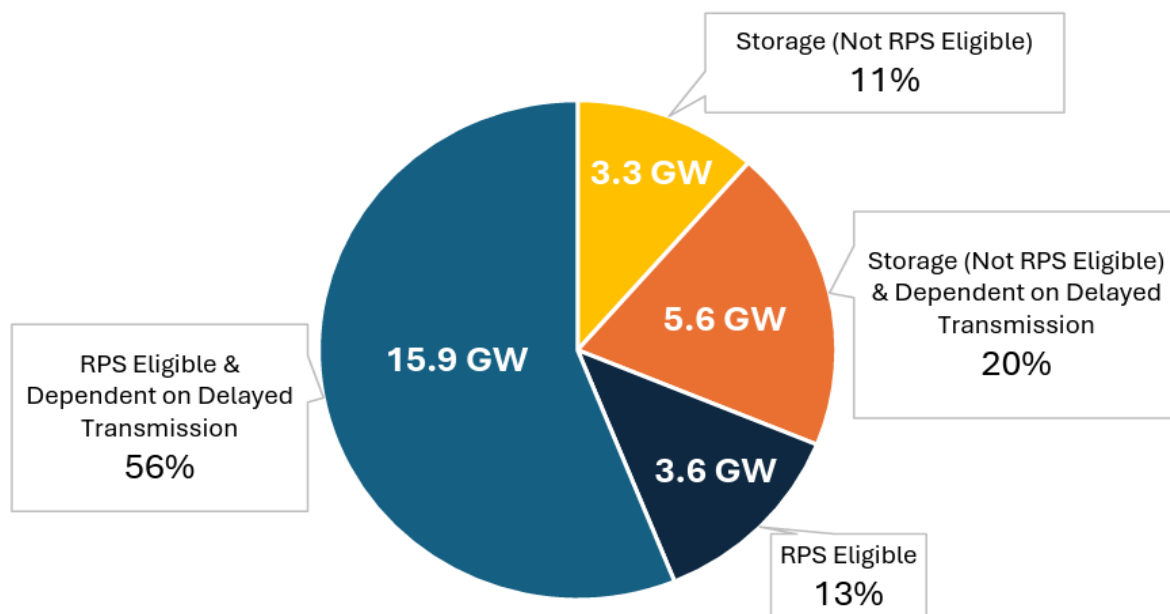


Figure 11. Delayed RPS-eligible vs non RPS-eligible resources (28.4 GW total).

Source: 2024 SB 1174 Data Request

To address these concerns, Senate Bill 1174 (SB 1174) requires utilities and CAISO to report in-service date changes and their causes. Table 1 presents a breakdown of delay causes reported in the CPUC’s 2024 SB 1174 findings.

Table 1. Types of delays impacting planned generation projects

In-service-date change reason	Transmission projects impacted	Dependent generation – RPS-eligible (GW)	Dependent generation – non-RPS-eligible storage (GW)	Total dependent generation – RPS-eligible & storage (GW)
Cust. Action	25	6.46	2.13	8.59
Other	26	4.71	1.06	5.77
Material	14	2.48	3.27	5.75
Clearances	5	3.68	0.15	3.83
Financing	9	1.14	2.21	3.35
Permitting (Utility)	8	1.85	1.18	3.04
Bundling Dependencies	13	2.68	0.09	2.77
Vendor Quality / Delays	3	1.9	0	1.9
Land Rights	6	1.51	0.03	1.55
Project Design	3	0.71	0.13	0.84
Permitting (Customer)	2	0.2	0	0.2
Workforce Availability	3	0	0.1	0.1
Supply-Chain Delays	2	0	0	0
Outage	1	0	0	0
Grand total	120	27.32	10.35	37.67

These figures highlight areas where policy interventions may be most impactful. For example, delays due to utility permitting, material shortages, and land rights disputes suggest that

interconnection policy should better account for real-world readiness screening projects for study. Furthermore, “customer action” delays, the single largest source, may signal issues with developer preparedness or speculative project behavior.

At the same time, CAISO’s efforts to manage deliverability allocations are becoming increasingly complex. As more developers seek full capacity deliverability (FCDS) to qualify for long-term contracts and resource adequacy eligibility, CAISO must conduct additional studies and manage stricter constraints across congested zones. In Cluster 14, the number of applications seeking FCDS increased significantly, while energy-only projects actually declined despite the cluster’s large size. This disproportionate shift suggests developers are increasingly aiming for higher value interconnection outcomes; but at the cost of a more intense workload and complexity of CAISO’s study process.

2.4 – Future Need and Current Decarbonization Trajectory

As of 2024, California had just under 90 GW of in-state generation capacity. Meeting the state’s energy goals through 2035 will require an estimated 7 GW of new capacity each year, a lofty goal. On paper, the current active IQ appears more than sufficient to meet that need, with over 106 GW of proposed capacity, and an additional 61 GW of new project proposals submitted under Cluster 15. However, as earlier sections have shown, the presence of projects in the queue is not equivalent to future operational capacity.

The tables below illustrate the fuel-type breakdown of both the active IQ and the Cluster 15 preliminary request list. A few key trends stand out:

- Standalone storage projects constitute the largest single category of queued capacity in both datasets.
- Hybrid solar-plus-storage is the next most common configuration, with fewer projects being submitted as solar-only resources.
- In Cluster 15, solar-only projects make up a slightly larger proportion than in the active queue, suggesting a modest resurgence of interest in simpler project designs.

Table 2. IQ Active Queue Interconnection Capacity in MW by Generation Type

Active Queue Types	Sum of net MW of type to grid
Combined Cycle	155
Storage	120
N/A	35
Gas Turbine	130.9
Storage	130.9
Hydro	5
n/a	5
Photovoltaic	23955.8
Storage	23202.3
n/a	753.5
Solar Thermal	100
Storage	100

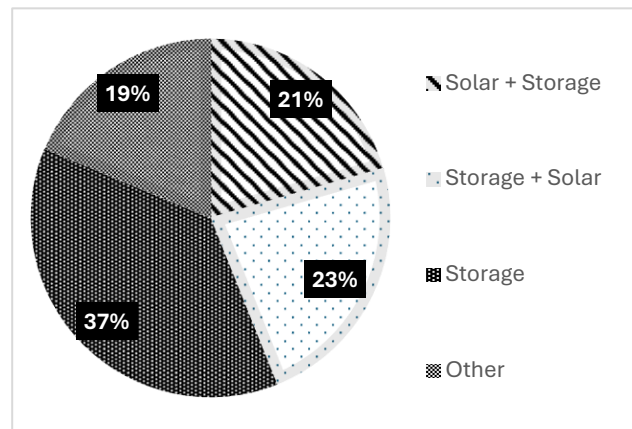


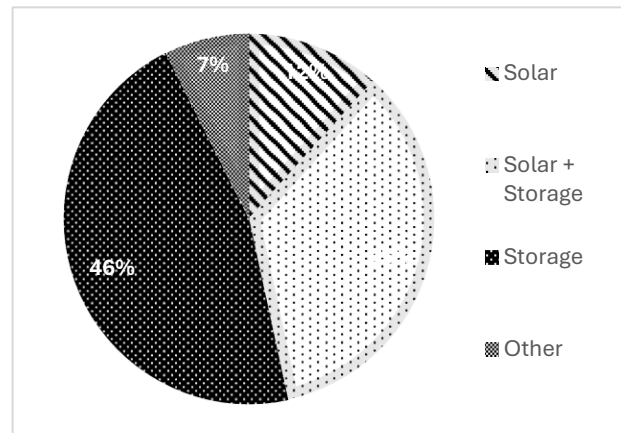
Figure 13. Visualization of Table 2

Active Queue Types	Sum of net MW of type to grid
Steam Turbine	389.28
Storage	336.28
n/a	53
Storage	73598.1
Other	500
Photovoltaic	26429.9
Storage	1400
Wind Turbine	3103.65
n/a	42164.6
Wind Turbine	7292.94
Photovoltaic	600
Storage	3037.37
(blank)	3655.57

Table 3. Cluster 15 Interconnection Requests in MW by Generation Type

Energy Generation/Storage Type	Sum of NET MW POI
Biomass/Biofuel	10

Energy Generation/Storage Type	Sum of NET MW POI
N/A	10
Hydro/Pumped- Storage Hydro	67.4
N/A	67.4
Photovoltaic/Solar	28454.16
N/A	7075.16
Storage/Battery	21179
Storage/Solar	200
Storage/Battery	31570.44
Combustion Turbine/Hydrogen	1649
N/A	27889.44
Photovoltaic/Solar	1870
Storage/Battery	162
Wind Turbine/Wind	804.6
N/A	404.6
Storage/Battery	400



If California is to meet its long-term clean energy goals, interconnection reforms must continue to ensure that the projects entering the queue reflect real commercial intent, technical feasibility,

and credible timelines. With storage and hybrid configurations leading the way, a streamlined and reliable IP will be key to unlocking their full potential.

2.5 – IPE 2021 Track 1: Final Proposal Posted 4/21/22

Track 1 of the 2021 IPE focused on near-term procedural adjustments intended to streamline the Cluster 14 study process and ensure CAISO could handle the record-breaking volume of interconnection requests. Rather than introducing major structural reforms, Track 1 implemented a series of **targeted updates** based on broad stakeholder consensus, deferring more contentious changes to future phases.

The key modifications made to the tariff in Track 1 included:

1. Revisions to Site Exclusivity Deposit Rules

For Cluster 14, site exclusivity deposits became **100% non-refundable** once projects entered Phase II studies.

For Cluster 15 and beyond, the **deposit amount was increased**, with **50% becoming non-refundable** after the initial scoping meeting.

Developers were also required to provide documentation of site exclusivity in order to proceed to Phase II.

2. Stricter Criteria for Emergency Generation Studies

Requirements were tightened to prevent unnecessary diversion of CAISO staff resources, reflecting the need to concentrate efforts on managing the workload associated with Cluster 14.

3. *Limits on Modifications to Parked Projects*

Projects that opted to "park" (pause progress while retaining a place in the queue) faced new restrictions. Modifications were limited to downsizing, fuel type changes, point of interconnection (POI) changes, or specific technology exemptions.

While these changes were modest and largely procedural, they signaled a shift in CAISO's posture. One from accommodating rapid growth to actively managing developer behavior. Track 1 served as a foundation for more substantive reforms proposed in Track 2, particularly those targeting queue discipline, site control, and deliverability allocation.

Interestingly, there was no immediate effect on withdrawal rates following Track 1 implementation, suggesting that the changes, while directionally important, did not materially alter project viability or developer behavior at the scale needed to reduce queue congestion.

2.5.1 – IPE 2021 Track 2: Final Proposal Posted 9/13/2022

Track 2 of the 2021 IPE process was much more significant in its impact compared to Track 1, focusing on foundational changes to the system that the CAISO believed would lead to a smaller queue and more manageable workload.

The changes made to the tariff in Track 2 included:

1. *Introduction of a minimum required term for PPAs to qualify for TPD allocation*

Previously, CAISO did not impose a minimum term length for PPAs used to qualify for

Transmission Plan Deliverability (TPD) allocation. Track 2 established a term requirement to ensure that only projects with credible long term commercial viability could access deliverability capacity. While public data does not confirm a direct cause, this policy shift likely stemmed from CAISO's internal recognition of project attrition tied to weak or short-term revenue arrangements. According to the withdrawn projects archive, nearly all withdrawals list the Interconnection Customer (IC) as the party requesting termination, a possible proxy for commercial infeasibility or insufficient financing.

2. *Acceptance of PPAs with non-LSE entities*

CAISO began allowing developers to submit five-year PPAs signed with non-load-serving entities (non-LSEs), provided that the offtaker demonstrates a resale contract with an LSE or posts a financial deposit (\$10,000/MW, minimum \$500,000). This offered developers more flexibility in securing deliverability while retaining safeguards to limit speculative entries. Stakeholder support for this provision was generally broader, as it opened new pathways for project financing without significantly increasing risk to the system.

3. *Reallocation of cost responsibility for low-voltage network upgrades*

For upgrades below 200 kV, each Participating TO may roll only up to 15 % of its low-voltage transmission net plant into rate base; any interconnection-driven costs above that cap must be financed by the Interconnection Customer with no cash reimbursement. The rule is intended to shield local ratepayers from disproportionate cost exposure and move

the ISO closer to participant-funding models used elsewhere, and it is unlikely that it had much if any influence on the state of the IQ.

4. Increasing Data Transparency

CAISO began publishing a precursor to the FERC ordered capacity availability heatmap in the form of a “Queue Report,” which includes relevant data about projects such as their PTO study area, sub-area, TPD allocation group and MW share, resource IDs, suspension/parking flags, and key phase-in dates. This decision was more proactive than the others in this IPE, providing developers additional tools to help assess the viability of a project. In the past, much of this information was difficult to obtain or out of date, leading to increased risk for developers and sometimes prompting the submission of multiple interconnection requests in multiple places to raise the likelihood of at least one being suitable for their needs.

5. Increased scrutiny of EO projects at risk of stagnation

While existing tariffs permit up to seven years between interconnection agreement and commercial operation date (COD), CAISO clarified its authority to terminate GIAs early based on quarterly reports showing lack of progress. The ISO signaled that it would begin invoking breach provisions more aggressively, particularly for EO projects showing no forward movement in permitting, financing, or construction. While termination still requires FERC approval, this change signaled a shift toward proactive queue management and tighter performance expectations.

These policy changes appear to have had a noticeable effect on developer behavior. In the months following the final proposal's publication in September 2022, the queue saw a sharp spike in withdrawals, with more than 120 projects exiting over an approximately 3 to 4 month span. While causality cannot be definitively assigned based on queue data alone, the timing suggests that Track 2 proposals may have played a role in accelerating project exits, particularly those lacking commercial readiness or facing financing challenges under the new criteria.

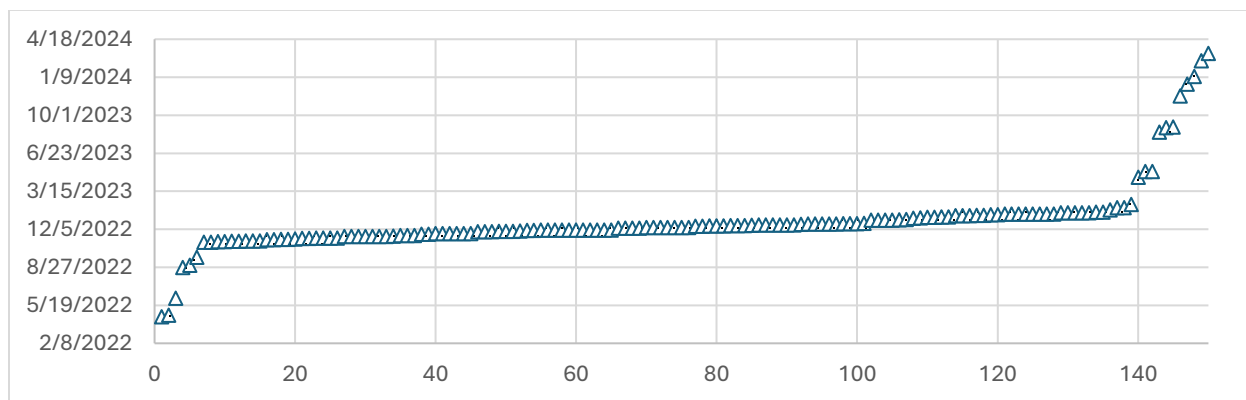


Figure 15. Visualization of withdrawals occurring after September 2023 final decision announcement

Taking a closer look at the stakeholder comments submitted during the IPE process offers insight into the priorities and concerns of various parties, including developers, regulatory agencies, and other stakeholders. While it may be intuitive that developers would push back against policies perceived to increase risk or restrict flexibility in the IP, a closer analysis reveals exactly why specific proposals generated concern and whether those concerns were ultimately borne out in practice.

Table 4. Stakeholder sentiment for minimum PPA Requirements

Sentiments	Count
Approves	6
Developer	2
LSE	3
Regulator	1
Somewhat Approves	1
Trade Association	1
Neutral / No Comment	8
Developer	3
LSE	3
Trade Association	2
Somewhat Oppose	2
Developer	1
Trade Association	1
Oppose	6
Developer	3
LSE	1
Trade Association	2

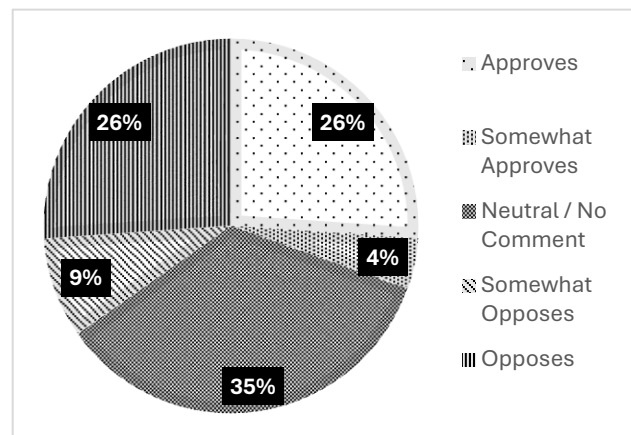


Figure 16. Visualization of Table 4

Table 5. Higher Fees, Deposits, & Other Criteria for Submitting an IR

Sentiments	Count
Approves	4
LSE	3
Regulator	1
Somewhat Approves	4
Developer	1
LSE	2
Trade Association	1
Neutral / No Comment	5
Developer	1
LSE	2
Trade Association	2
Somewhat Oppose	6
Developer	4
Trade Association	2
Oppose	4
Developer	3
Trade Association	1

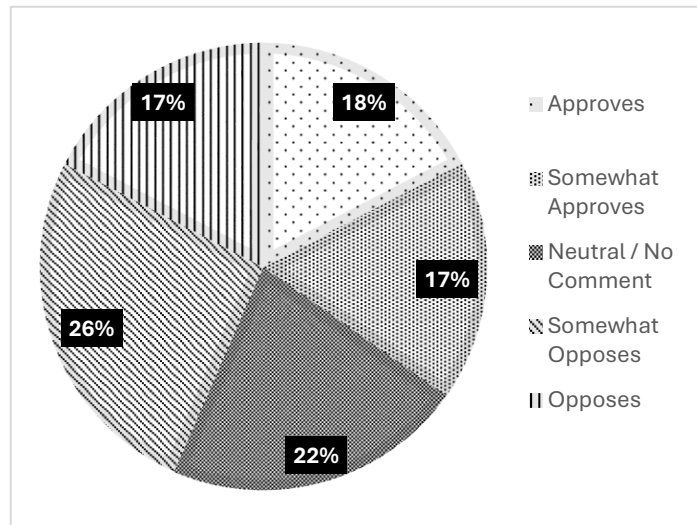


Figure 17. Visualization of Table 5

2.6 – IPE 2023 Track 1: Final proposal posted 04/13/2023

Much smaller than any of the other decisions made by the ISO, Track 1 of IPE 2023 cleared calendar time to allow for a more robust set of proposals to be put forward for Track 2 of the IPE which would be compiled throughout the rest of the year.

Changes made include

1. Pausing Cluster 15 and postponing Cluster 14 Phase 2 Study deadline

Provide a grace period for interconnection customers to withdraw without penalty any time prior to April 1st, 2024, while also postponing Cluster 14 Phase 2 deadline to January 31st, 2024.

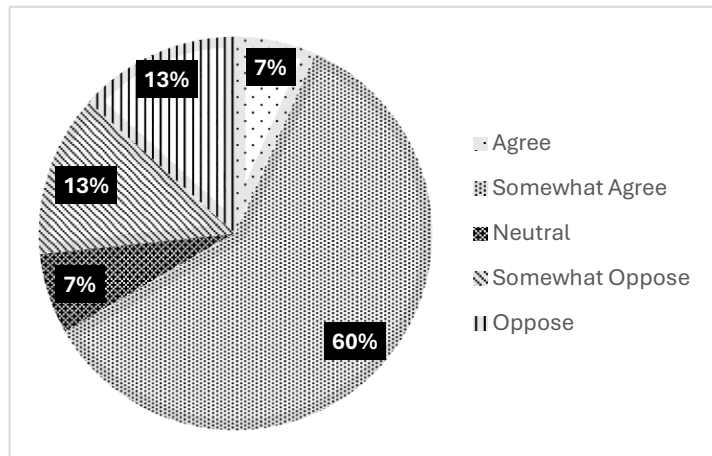
These decisions were largely procedural and did not result in a noticeable change in project withdrawal rates, as expected given their non-restrictive and optional nature. Unlike other IPE tracks, Track 1 of 2023 did not impose new requirements or modify the interconnection framework in any significant way.

Stakeholder engagement for this track was limited, with only one round of comments solicited. Nearly all respondents agreed that urgent steps were necessary to manage the unprecedented volume of Cluster 14 interconnection requests. The lack of opposition underscores the consensus view that this delay was a necessary short-term intervention while longer-term reforms were being formulated.

Table 6. Pausing Cluster 15 and postponing Cluster 14 Phase 2

Sentiments	Count
Agree	1

LSE	1
Neutral	1
Developer	1
Oppose	2
Developer	1
Trade Association	1
Somewhat Agree	9
Developer	5
Trade Association	3
Trade Association / LSEs	1
Somewhat Oppose	2
Developer	1
Trade Association	1
Grand Total	15



2.6.1 – IPE 2023 Track 2: Final proposal posted 03/29/2024

Track 2 of the 2023 IPE continued the trend of making significant changes to the existing IPEs initiated by Track 2 of the 2021 IPE, most significantly modifying the methodology used for determining the amount of available capacity in different geographic study areas. Projects applying for FCDS in TPD zones will now be assessed and scored to determine which projects are eligible for study.

This change reflects a growing urgency within CAISO to screen out projects unlikely to secure transmission deliverability and streamline the IP by preemptively filtering out less viable proposals. In doing so, CAISO aims to ensure that study resources are focused on the subset of projects most likely to advance to commercial operation.

The Zonal Approach

A central tenant of the track 2 initiative was to distinguish transmission constraints by rough geographical areas, or Transmission Planning Deliverability (TPD) zones, calculating the total available and planned deliverability that transmission infrastructure located in each zone can accommodate. From cluster 15 onwards, projects submitting interconnection requests will be scored based on criteria such as commercial interest, system need, and project viability indicators.

Projects are then ranked in descending order of score. Once the cumulative capacity of ranked projects reaches 150% of the zone's deliverability capacity, no further requests are studied in that zone. The overage is allowed to account for potential attrition or project withdrawals, reducing the risk of unused deliverability.

This marks a significant departure from the previous approach, which did not impose hard thresholds or directly limit how many projects could apply in a given area. By introducing a ranking-based cap, CAISO is not only limiting the number of projects studied in constrained zones, but also shifting the IP closer to a competitive resource procurement model, where developer readiness and system alignment become prerequisites for advancement.

While stakeholder reception to the zonal scoring system was mixed, CAISO justified the change as necessary to relieve systemic study overload and refocus efforts on projects that meet both commercial and planning benchmarks.

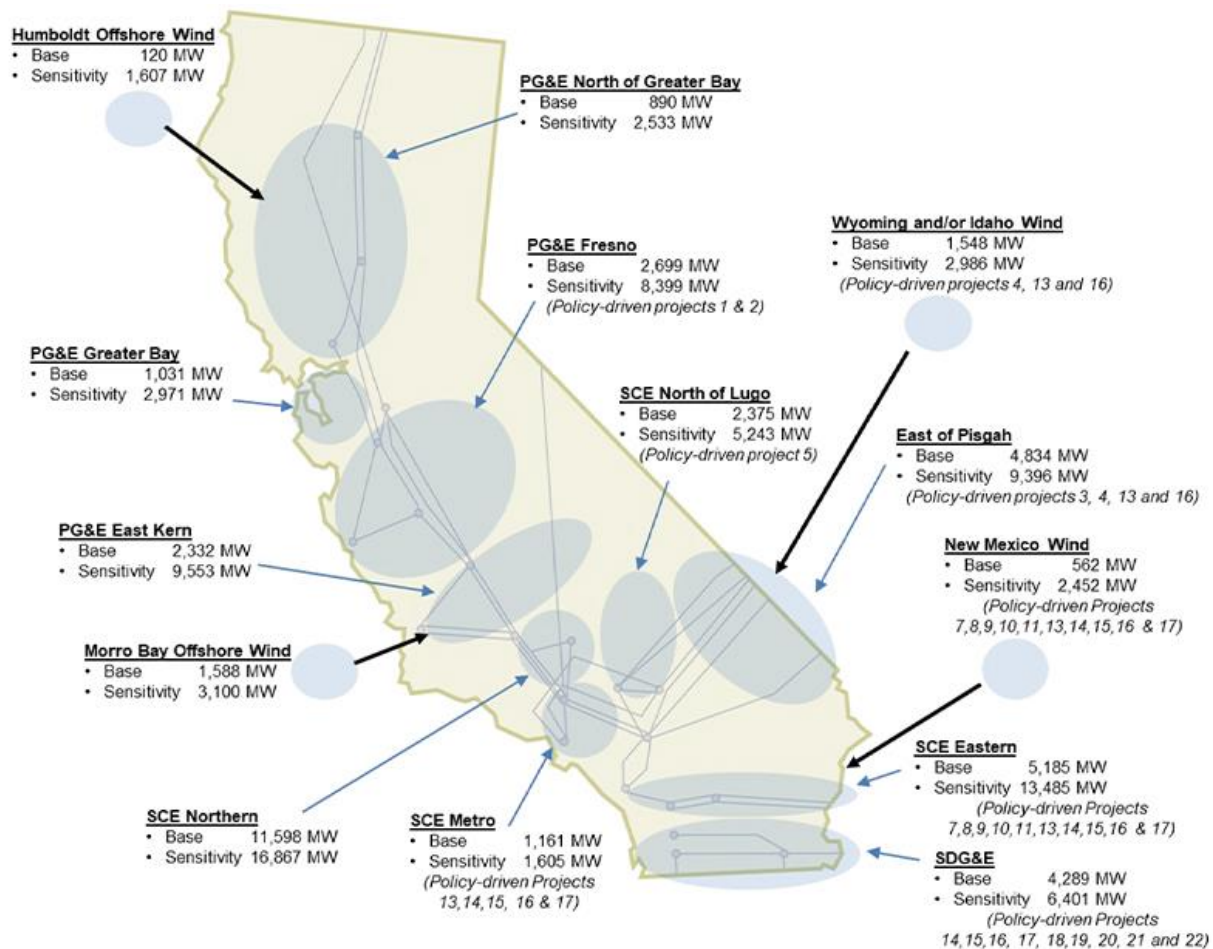


Figure 19. CAISO published map depicting the new interconnection zones

The criteria used to score interconnection requests under the new zonal approach are summarized in the table below. Each project is evaluated across three major categories: commercial interest, project viability, and system need, for a total of 100 points.

Table 7. New CAISO proposed TPD Allocation Scoring Criteria

Main Category	Max Points	Subcategory	Max Subcategory Points	Description
1. Commercial Interest	30 points	a. LSE Allocation	30 points	Points based on the proportion of project capacity allocated by Load Serving Entities (LSEs) (e.g., via executed or pending PPAs, or other demonstration of interest). Full allocation = 30 points.
		b. Non-LSE Interest	25 points (max)	Points for signed affidavits from unaffiliated non-LSE entities (e.g., corporate buyers) committing interest. Only one non-LSE entity per project per cycle is allowed.
2. Project Viability	35 points	a. Engineering Design Completeness	15 points	Points for detailed, complete, and plausible engineering design plans for generation and interconnection facilities.
		b. Development Progress	10 points	Points for demonstrated progress on permitting, CEQA, land control, etc.
		c. Existing Facility Affiliation	10 points	Points if the project is an expansion or repowering of an existing interconnection or generation facility.
3. System Need	35 points	a. Location in Local RA Area	15 points	Points for being located in an area of high local reliability need (Local RA area).
		b. Long Lead-Time Resource Type	10 points	Points for resource types identified by the CPUC as “long lead-time” (e.g., offshore wind, geothermal).
		c. Alignment with IRP Resource Types	10 points	Points for matching resource types and timing identified in CPUC’s Preferred System Plan (PSP) or RSP.

These criteria are intended to pre-screen projects and reduce queue bloat by favoring proposals with stronger commercial backing, more complete development progress, and alignment with identified system needs. But how have developers responded to this new structure?

A closer look at withdrawal patterns in the Cluster 15 queue provides some insight.

The maps below compare the geographic distribution of withdrawn projects and active projects in Cluster 15, visualized by their selected Points of Interconnection (POIs). The left map displays total MW of generation from projects that withdrew between October and May, while the right map shows generation capacity still listed in the Cluster 15 queue as of May 1, 2025. In both maps, pie charts summarize the transmission capacity headroom at the POIs selected by these projects. Red slices represent projects targeting POIs with no available capacity (according to CAISO's deliverability heatmap), while blue slices represent projects located at POIs with remaining TPD headroom.

Over one-third of withdrawn projects had selected POIs where no deliverability remained, suggesting the CAISO heatmap may have influenced developers' decision to withdraw. While developers may have had access to this information via other channels, the public release of this spatial data likely provided many with their first clear signal that their chosen POI was either oversubscribed or contested. Encouragingly, this effect seems to be working as intended. Among newly submitted Cluster 15 requests, the proportion of projects targeting oversubscribed POIs has declined to just under half, indicating that developers are responding to clearer locational guidance and adjusting their proposals accordingly.

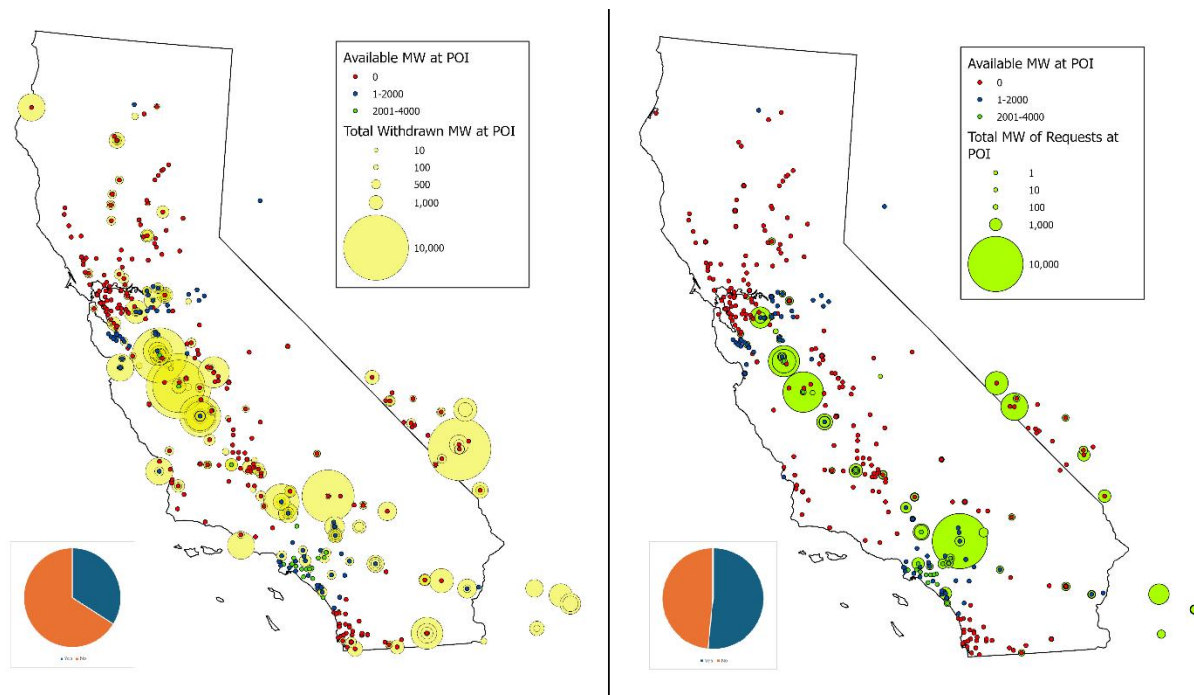


Figure 20. Total MW of withdrawn vs non-withdrawn Cluster 15 projects by POI; pie chart shows TPD headroom at POIs for each category

While developers may have become aware of transmission constraints through various channels, it is likely that CAISO's public deliverability heatmap served as a key tool in informing many that their proposed POIs lacked sufficient headroom. This added transparency may have directly influenced developers' decisions to withdraw or revise their interconnection requests.

With this change, CAISO has established clearer guiderails for project siting and submission, aiming to reduce speculative proposals and queue congestion. Among the roughly 340 projects that withdrew from the Cluster 15 queue, over one-third had identified POIs with no remaining TPD headroom. Notably, more recent submissions to Cluster 15 show a decrease in this ratio, with fewer projects now targeting oversubscribed POIs. While it is too early to say with certainty, the data suggests that developers are adjusting their behavior in response to the new zonal framework and the public availability of capacity data.

Other changes made include:

1. Merchant-Option Zones

This new designation allows projects wishing to interconnect to POI's with less than 50 MW of available TPD to skip the scoring process by posting a \$10000/MW deposit, self-fund all network upgrades, and forgo any reliability network upgrade reimbursement opportunities. While an expensive option, forgoing the cluster studies may be an attractive enough proposal for some developers to consider it.

2. Modified pathways for energy-only projects

A 150% ceiling of energy-only need designated by the CPUC was also established for developers seeking to connect their projects as energy-only while also receiving a reliability network upgrade (RNU) reimbursement. Energy-only projects willing to forgo this RNU reimbursement would be able to skip the scoring process and interconnect anywhere.

3. Stricter time in queue limits

The ISO defined more clearly the stricter scrutiny that it would be applying towards stalled projects, requiring milestones such as an executed GIA and 3rd security deposit posted in the event of three cumulative COD extensions.

4. Earlier financial postings for shared upgrade costs

When the first project on any shared upgrade has posted its 3rd security deposit or provided a notice to proceed, CAISO will notify all other funders of the upgrade to either post their financial contributions within 9 days or be withdrawn.

5. Scoring-Based Access for FCDS Projects in TPD Zones

In TPD option zones, (areas with at least 50 MW of available or planned deliverability) FCDS applicants will be scored and ranked based on criteria such as site control, executed PPAs, LSE endorsements, and alignment with planning goals. Only the top-scoring projects, representing no more than 150% of available deliverability, will be included in the study process.

Much like the previous IPE 2021 Track 2, this proposal marks a structural shift in how CAISO screens, scores, and advances projects through the IP, tightening existing requirements for projects to remain in the IQ, as well as introducing further barriers to being considered for study in the first place.

Finally, the proposal formalizes the publication of a Transmission Deliverability Availability Heatmap, in compliance with FERC Order 2023. It also expands the scope of publicly available queue data, improving developer visibility into siting feasibility and study prioritization criteria.

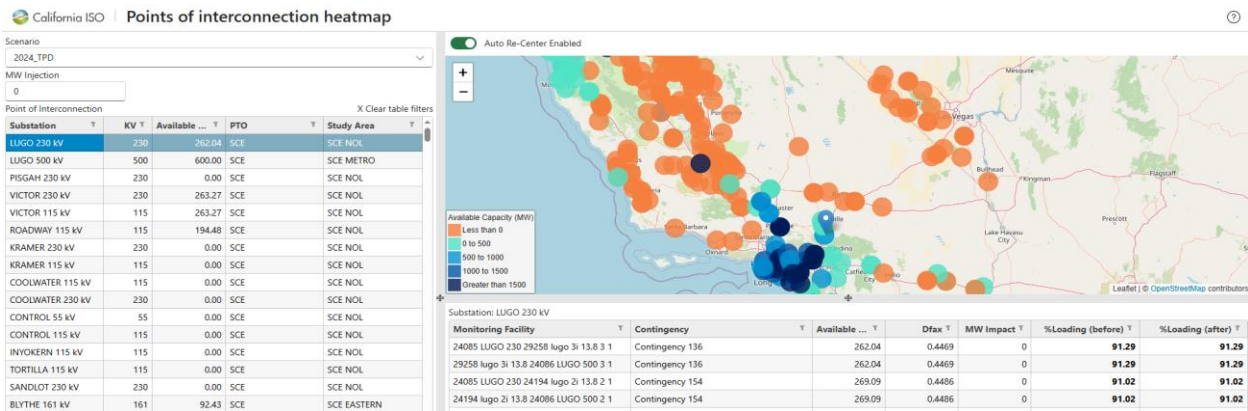


Figure 21. CAISO published heatmap of TPD availability

Stakeholder participation in this track of the IPE was much greater than the first, with the final draft proposal coming only after several in depth rounds of workshoping with stakeholders via soliciting of comments and online discussions. Despite this active participation however, many developers did not exhibit full confidence in the comments left in response to the final draft publication IPE 2023 Track 2.

Table 8. Sentiment for new TPD Allocation structure

Sentiments	Count
Somewhat Agree	8
Developer	2
LSE	4
Regulator	1
Trade Association	1
Neutral	1
Trade Association	1

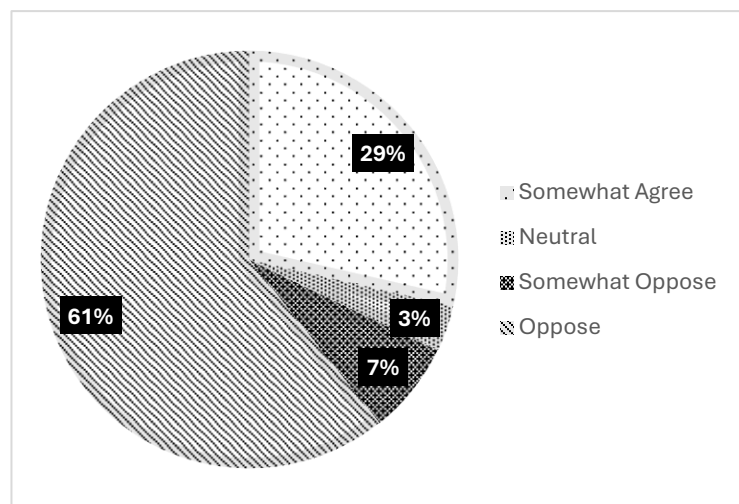


Figure 22. Visualization of Table 8

Somewhat Oppose	2
Developer	1
Trade Association	1
Oppose	17
Developer	14
Trade Association	3

Independent developers and other non-LSE stakeholders expressed strong reservations about the proposed scoring framework, particularly its emphasis on load-serving entity (LSE) engagement. Under the new system, the “Commercial Interest” category accounts for up to 30% of a project’s total score. As drafted, a single LSE could allocate points toward a small number of large, utility-sponsored projects or prioritize a limited subset of projects totaling 25% of its transmission-capacity allocation. Developers argued that this would disadvantage unaffiliated or newer entrants, potentially excluding viable projects solely due to the absence of an incumbent counterparty. Several comments cited this as raising concerns under the non-discrimination principles outlined in FERC Order 888.

Beyond LSE influence, developers highlighted multiple implementation challenges. Several respondents described the proposed submission timeline as too short to incorporate new transmission constraint data, renegotiate site control, and prepare the required 50% engineering design documentation. There were also repeated concerns about vague or inconsistent terminology within the draft, particularly in how the proposal used terms like “zone” versus “constraint area,” leading to confusion about the geographic applicability of scoring thresholds.

Finally, some stakeholders expressed frustration with how the scoring system might favor certain resource types. The 100-point bonus proposed for long-lead-time resources like offshore wind or geothermal was seen as potentially crowding out solar-plus-storage portfolios that can be built faster but may struggle to compete in the ranking process. Others cautioned that continued inclusion of large volumes of energy-only (EO) projects which are ineligible for RPS deliverability would continue to burden CAISO's study resources without helping meet procurement targets.

2.6.2 – IPE 2023 Track 3: Final proposal posted 02/25/2025

Track 3 of the 2023 IPE closed this iteration with a set of focused refinements aimed at improving the efficiency and effectiveness of the study process. While smaller in scope than the earlier tracks, these changes target key operational bottlenecks and developer decision-making dynamics.

1. Intra-cluster prioritization to make use of existing short-circuit headroom

To maximize projects moving through the queue as quickly as possible while it remains in its crowded state, CAISO elected to award certain projects awaiting deliverability upgrades to energize up to the point of triggering long-lead time RNU's.

2. TPD Allocation Overhaul and Elimination of "Parked" Projects

Replacing the previously utilized A, B, C, D group structure for transmission plan deliverability (TPD) allocation with a 3 group system composed of a PPA group, a Commercial Operation group, and a Conditional group. This decision reiterated the

importance of having a PPA to prospective ICs, as well as removing the group that allowed shortlisted projects to park for longer than intended.

3. *Postpone the deadline for the second interconnection financial security posting to allow developers to know their deliverability status before further committing.*

This would allow developers additional information and time to make their decisions, hopefully leading to a more optimized queue where developers self-remove themselves when the risks of not meeting operational timelines is made apparent.

4. *Clarification of Reserved TPD Headroom for Priority Resource Types*

Responding to recurring stakeholder questions, CAISO clarified its authority to reserve deliverability identified in the Transmission Planning Process (TPP) for select long-lead-time or resource adequacy-critical technologies. These include offshore wind, pumped hydro storage, and other resources designated as Local Reliability Area (LRA) priorities.

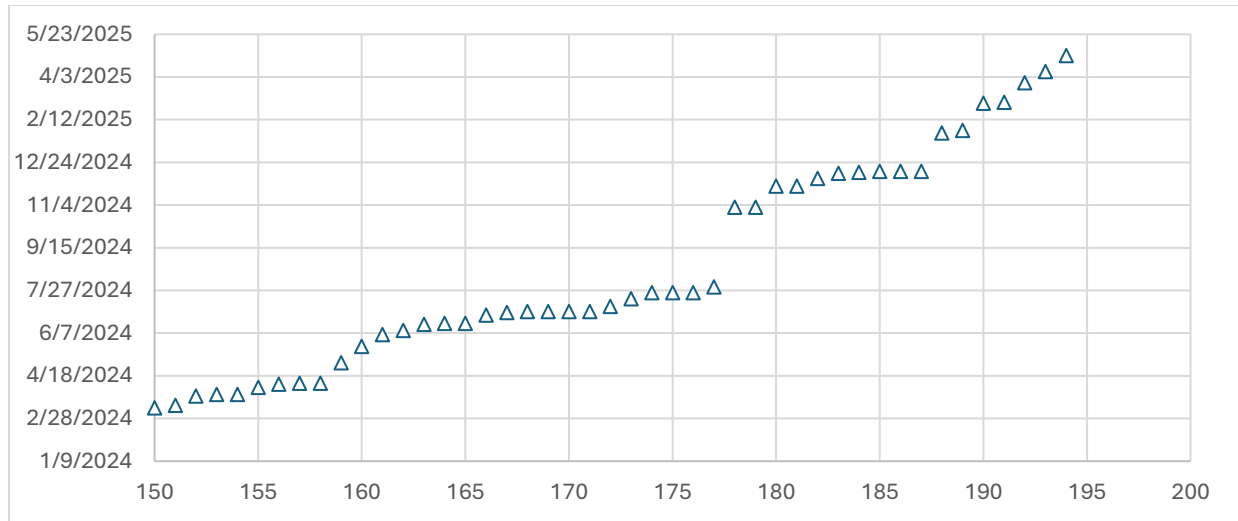


Figure 23. Close-up of project withdrawals between 2024 and 2025; small clusters emerge in late 2024, loosely coinciding with the start of IPE 2023 Track 3

Table 9. Developer sentiment regarding changes to parked project standards

Sentiments	Count
Agree	1
Regulator	1
Somewhat Agree	5
Trade Association	1
Developer	1
LSE	3
Neutral	6
Trade Association	1
Developer	2
LSE	1

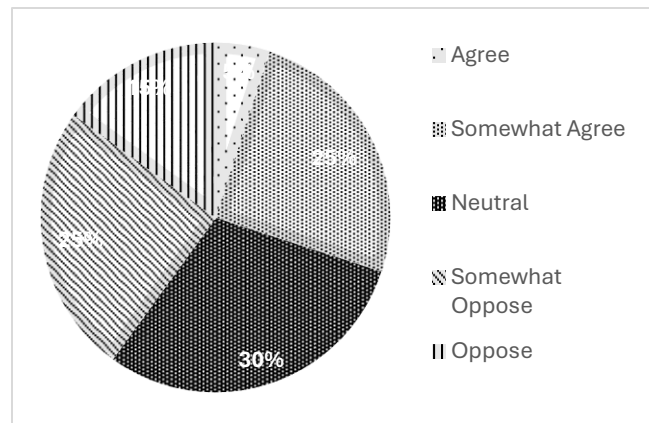


Figure 24. Visualization of Table 9

Sentiments	Count
Trade Association (NGO)	1
Developer	1
Somewhat Oppose	5
Trade Association	1
Developer	4
Oppose	3
Trade Association	1
Developer	2

As transmission plan deliverability (TPD) becomes increasingly valuable, so too does the influence of Load Serving Entities (LSEs) in determining which projects ultimately succeed. Developers have voiced growing concern over the extent of this influence and whether there will be any oversight mechanisms in place to ensure that access to the queue remains fair and competitive for all participants.

The ever-tightening restrictions being placed on current and future projects was also noted by developers, with many hoping that CAISO would reconsider allowing EO projects to eventually achieve FCDS despite not entering the queue requesting that deliverability status. Many projects face uncertainty early in the development process, and developers are worried that CAISO would be unnecessarily barring otherwise viable EO projects to eventually obtain full deliverability. For now, it seems that while most understand the position CAISO is in, they are becoming more and

more apprehensive of the increased risk associated with LSEs ability to dictate a project's ability to be studied, as well as awarded TPD.

2.7 – Non-CAISO Entity Policy Initiatives

While the most granular changes to the IP are decided and implemented by CAISO, broader policies dictating factors such as governmental procurement of renewable generation or new federal guidelines that introduce new guidelines and systems must be considered when discussing CAISO efforts.

CPUC: Reliable and Clean Power Procurement Program (RCPPP)

The CPUC has recognized through their research that the uncertainty that has been introduced due to a lack of a standardized procurement program to help guide the development of clean generation in California is a pressing matter. Though still in the proposal stage, the program has the potential to offer developers clearer signals of commercial interest backed by government procurement. The authors hope that this will also result in an increased percentage of projects that are able to secure eligible PPAs earlier on in the development process, providing increased certainty to developers that their projects will be able to proceed through the IP.

While this may be able to help developers obtain the contracts necessary to participate in the study process and be assigned deliverability, it is important to note that nothing in the current proposal will guarantee deliverability for projects that obtain PPAs via RCPMP guidance. As it is directly related to the deliverability process, the CPUC recognizes that such a guarantee of

deliverability would have to be a result of a CAISO policy that goes through its own creation and refinement process.

FERC Order 2023

In the midst of the 2023 IPE reform drafting process, FERC released a new set of guidelines for the interconnection tariffs of systems operators in the USA. CAISO, as an early adopter of the cluster process, deliverability allocation, and site-control obligation, did not have to modify their tariff to the extent that other ISOs around the country required. Most of the modifications made were procedural or terminological in nature, with some restrictions being made more strict and others remaining unchanged due to CAISO already requiring stricter standards than the FERC requirement. CAISO submitted its compliance proposal to FERC shortly after the publication of their new guidelines, integrating the changes into the ongoing Cluster 15 study process.

Summary

A closer look at project withdrawals over the past year suggests that the IQ is beginning to stabilize. Although the period is relatively short and there still exist sporadic clusters of withdrawal, observed values suggest that the projects that were least likely to meet commercial operation have vacated their position in the queue, and those that remain are of higher confidence of success. While there are still further refinements to be made to both the IP as well as the TPD allocation structure, CAISO is taking aggressive and decisive action to reduce the IQ down to more manageable levels. This can be seen in their stricter requirements for valid PPA's, clear

150% MW capacity ceiling on studies being submitted with certain points of interconnection (POI), and efforts in removing stalled projects from the queue.

Not only do these reduce the number of active projects currently in the queue but also reduce the number of study requests for future clusters as developers better understand what is and is not required for a viable project. The rapidly shrinking list of projects on the preliminary Cluster 15 study list is an indication that these new requirements are achieving their objectives, but it will be important for CAISO to remain vigilant of potentially adverse outcomes. While RA obligation and greater state renewable energy goals are still being met, it is a common sentiment among developers in the stakeholder feedback that the increasing difficulty of doing business in California may eventually lead to a critical mass of developers choosing to develop their projects in other, less policy intensive regions. While we are on track to meet our statutory 2030 renewable generation milestones, build-out will need to be accelerated further throughout the 2030's to achieve 2045 goals.

CAISO plays a relatively unique and involved role as a systems operator, acting closely in tandem with the CPUC and the CEC to help achieve both the states renewable energy goals as well as its own of maintaining a balanced and operational energy grid for its customers. While it allows for a greater ability for CAISO to plan with LSEs and state entities, it has necessitated greater and greater involvement of LSE's in determining which projects will enter the queue and how likely they are to eventually be awarded with deliverability allocation. In the next section we will look through interview responses and commentary left by stakeholders to better understand the viability of the changes being made, and what knock-on effects they may have on developers.

Chapter 3: Interview Analysis

A central component of this analysis involved interviews with various stakeholders in the IP, including developers, regulators, and LSEs. By combining the sentiments, recommendations, and other input found in comments solicited by regulatory organizations with the real-world challenges these groups face in their day-to-day operations, a clearer picture emerges of how the IP is functioning compared to past iterations.

3.1 – Stakeholder Types

Interview participants can broadly be categorized into three groups: developers, load-serving entities, and regulators. Each plays a distinct role in the IP and brings unique perspectives shaped by their objectives and operational constraints.

Developers – Developers are private or corporate entities that construct generation facilities with the intent of selling energy into the market. They represent the majority of active participants in the IP and are most directly affected by changes to CAISO’s interconnection procedures. Their ability to move projects forward is shaped directly by policy decisions regarding the interconnection process and the administration of it.

A consistent theme among developers interviewed is the desire for greater predictability and transparency in the IP. Frequent policy shifts, whether from CAISO or the CPUC, or any other agency for that matter, are viewed as disruptive; Especially when they raise entry barriers

through increased deposit requirements, site control documentation, or commercial readiness standards. Although California remains an attractive market due to high renewable energy demand and ambitious state targets, several developers expressed frustration at the regulatory instability and lack of clear long-term planning signals.

Many also raised concerns about equity and market concentration. The new scoring criteria introduced in the IPE reforms emphasize factors like detailed engineering design, demonstrated commercial interest from LSEs, established site control, all factors that implicitly favor larger, well-capitalized firms. Smaller or early-stage developers often lack the resources to satisfy these thresholds early in the development cycle which can potentially locking them out of meaningful participation. Several stakeholders warned that this could lead to reduced competition and a narrower pool of project sponsors, with long-term consequences for innovation and cost-effectiveness.

Developers acknowledged the need for stronger queue management but questioned whether CAISO's current approach of increasingly tightening barriers for entry to and progress through the queue is the most effective path. A more adaptive framework that filters speculative projects without sidelining smaller developers was frequently cited as a preferable alternative.

The following table synthesizes interview responses, highlighting areas of agreement and tension across key policy domains.

Table 10. Developer sentiment regarding possible improvements to IC process

Aspect	Positive (Pros)	Neutral	Negative (Cons)
Interconnection Management	Appreciate initial feasibility assessments (fatal flaw analyses); improved CAISO heat maps due to regulatory requirements.	Accept complexity as inherent in processes.	Significant queue congestion and inefficiencies; frustration with slow queue clearing; speculative projects causing delays.
Deliverability Allocation	Understand importance of deliverability to market participation; supportive of structured procurement frameworks (RCPPP).	Recognize regulatory frameworks as necessary.	Deep frustration over deliverability uncertainty and complexity; difficulty obtaining PPAs without guaranteed deliverability.
Transparency	Positive view of incremental transparency improvements (heat maps).	Recognize data complexity and the competitive advantage issue from full transparency.	Critical of data accuracy, granularity, and timeliness; frustration over inadequate GIS visualization tools.
Regulatory Stability	Appreciate CAISO stakeholder engagement and responsiveness.	Acknowledge inevitability of some regulatory uncertainty due to rapid market changes.	Concerned about frequent changes and instability in regulatory rules affecting planning and financial decisions.

Policy Regulators – These entities are responsible for designing and implementing policies that govern the IP. Their decisions are guided by California’s renewable energy mandates and have significant downstream impacts on both developers and LSEs. Regulatory representatives consistently emphasized the urgency of reform, noting that the current IQ is excessively congested and poses risks to timely resource integration. However, they expressed confidence in

the iterative stakeholder engagement process as an effective means to address these challenges systematically.

Regulatory interviewees acknowledged the persistent challenges posed by the current queue congestion and the difficulty of implementing reforms at the pace required by evolving energy targets. Still, they emphasized that CAISO and CPUC have developed a strong collaborative framework, particularly around recent initiatives like the IPE and busbar mapping. These reforms were described as both necessary and overdue, even if difficult to implement efficiently in practice.

While acknowledging the operational constraints developers face, regulators understand the need for proactive intervention given the rate of change. One policy cited as being relatively contentious was the reservation of transmission deliverability for offshore wind in the Humboldt Bay study area. Developers questioned the decision to allocate this quantity of capacity to resources still in early development stages while commercially ready projects remain without deliverability. Regulators responded that such decisions reflect long-term system planning priorities and emphasized that areas where policy has misaligned with infrastructure realities have been identified through internal reviews and formal stakeholder feedback for future refinement.

Finally, regulators pointed to growing coordination between policy instruments such as the IRP and the Reliable and Clean Power Procurement Program (RCPPP) as evidence of policy alignment. However, they also admitted that aligning these theoretical models with on-the-ground transmission constraints remains a challenge. As one regulator put it, “There’s a disconnect between what looks optimal on paper and what is feasible from an engineering

standpoint.” Ensuring that planning frameworks better reflect transmission realities remains a shared priority for future reform cycles.

Table 11. Policy Regulator sentiment regarding possible improvements to IC process

Aspect	Positive (Pros)	Neutral	Negative (Cons)
Interconnection Management	Effective collaborative relationships (CAISO-CPUC); necessary reforms being made.	Accept complexity of managing queue as natural.	Persistent challenges due to queue congestion, speculative projects, and difficulty implementing efficient reforms.
Deliverability Allocation	Supportive of clearly defined deliverability allocation process; strategic planning (busbar mapping, RCPMP).	Accept role as balancing market reliability and affordability.	Concerned about tensions between market reliability demands (CAISO) and affordability pressures (CPUC).
Transparency & Communication	Incremental transparency improvements (heat maps, portals, stakeholder feedback mechanisms).	Accept that transparency improvements are gradual.	Recognize existing data sharing inadequacies; potential friction due to confidentiality requirements.
Policy Alignment & Collaboration	Strong alignment between CAISO and CPUC policies (IRP, RPS, RA, RCPMP); supportive of strategic policy frameworks.	Recognize complexities due to overlapping jurisdictions (FERC, CPUC, CAISO).	Frustration with difficulty aligning theoretical planning models (RESOLVE, IRP) with practical transmission realities.

Load Serving Entities – LSEs serve as the offtakers for generation projects and are responsible for both delivering this energy to their customers as well as ensuring that their procurement portfolios meet regulatory mandates established by the CPUC. These mandates include, but are

not limited to the Renewable Portfolio Standard (RPS) and Resource Adequacy (RA) obligations. Operating under strict oversight, LSEs must balance affordability, reliability, and emissions requirements while coordinating with developers and regulators throughout the IP.

Policy changes introduced through IPE 2021 and IPE 2023 expanded the role of LSEs, granting them greater influence over which projects are selected for study through deliverability scoring frameworks. While this authority allows LSEs to help shape the resource mix more directly, it has also introduced operational strain. LSE representatives indicated that the expanded responsibilities around project scoring and documentation review have added considerable workload without a proportional increase in resources or staffing.

Despite their distinct position in the IP, LSEs share several concerns with developers. LSE interviewee expressed frustration over the difficulty in obtaining basic information about queued projects, including contact details, company affiliations, and project viability. In some cases, attempts to engage with developers regarding potential procurement opportunities were hindered by the absence of publicly accessible project data.

This lack of reciprocal transparency complicates efforts to meet procurement targets efficiently and raises questions about how well aligned the current IP is with the practical needs of load-serving entities. While LSEs support reforms that help prioritize viable, policy-aligned projects, they also caution that added responsibility must be matched by better tools, clearer communication channels, and greater visibility into the project landscape.

Table 12. LSE sentiment regarding possible improvements to IC process

Aspect	Positive (Pros)	Neutral	Negative (Cons)
IQ & LSE Points System	Supportive of LSE points system as practical solution to winnow the queue; allowed clear expression of LSE procurement preferences.	Accept role as necessary despite increased responsibility.	Concerns about burden of cost and effort on LSEs to manage scoring process; insufficient queue reduction from implemented measures.
Resource Preferences	Clear preference for co-located generation plus storage due to reliability and energy sufficiency considerations.	Understand developers' different perspectives on resource preferences.	Frustration with mismatches between developer proposals (stand-alone batteries/solar) and LSE reliability needs.
Transparency & Information Sharing	Strong advocate for improved transparency (contact info, standardized naming, better mapping/GIS data).	Recognize competitive confidentiality concerns from developers.	Critical of inadequate data sharing practices, making procurement decisions unnecessarily difficult and costly.
Regulatory & Economic Stability	Supportive of clear regulatory frameworks (IRP, RCPPP) guiding procurement and project assessment.	Recognize complexities inherent in navigating CPUC and CAISO regulations.	Significant concerns about rapidly shifting regulatory rules causing planning instability and uncertainty.
Transmission & Infrastructure Planning	Supportive of strategic planning via CPUC's TPP and IRP processes for clear infrastructure development.	Recognize inevitable complexity in aligning procurement with transmission planning.	Deep concern over mismatches between CPUC planning (IRP/RESOLVE models) and actual queued resources; concerned about high-cost transmission upgrades and realistic feasibility of projects (offshore wind, Wyoming wind).

3.2 – Common Sentiments and Themes

Where relevant, themes identified in interviews are interpreted in light of related stakeholder comments, policy documents, and procedural reforms to provide a more integrated picture

Transparency and Mutual Disclosure of Information

Across all interviews, participants consistently emphasized the importance of improving access to information within the IP. While opinions varied regarding what information should be made available and to whom, there was broad consensus that enhanced transparency could significantly improve queue management and inter-stakeholder coordination. Several interviewees pointed out that current information asymmetries contribute to inefficiencies, delays, and unnecessary speculation.

Developers expressed concern that disclosing too much about their projects could compromise their competitive advantage. This tension is especially apparent as LSEs are now granted more authority in selecting projects for study under CAISO's reformed scoring structure. At the same time, LSEs reported difficulty accessing even basic information about proposed projects, including points of contact, project status, or ownership. One LSE representative described being unable to obtain sufficient information about a project already in the queue, despite being tasked with evaluating its viability for procurement.

These concerns reflect a broader misalignment in objectives and vision. While developers prefer minimal disclosure, LSEs are now expected to act on project-level information that may not be readily accessible. Without a mechanism to improve data availability, the risk increases that project selection will be based on incomplete or inconsistent information. Though developers are justified in seeking to protect sensitive data, a growing number of stakeholders (including

CAISO) recognize that some level of structured disclosure is necessary to ensure LSEs can perform their expanded role effectively.

However, CAISO's ability to mandate disclosure is limited under current tariff language. Any meaningful reform would likely require further rule changes or FERC-approved tariff modifications. While FERC Order 2023 introduces new expectations around data transparency, its impact remains limited in scope and early in implementation. Voluntary transparency encouraged through enhanced developer-LSE communication or standardized public reporting may represent an interim solution while longer-term reforms are pursued.

Deliverability

Another topic that generated considerable discussion among stakeholders is the allocation of transmission deliverability and how CAISO intends to refine this policy in future iterations of the IPE. Under the most recent reforms, LSEs are now responsible for both allocating deliverability among generation projects interconnecting within their service areas and contributing to the assessment of interconnection requests used to determine which projects are included in CAISO's study process. While developers did not express significant concern that LSEs would unduly favor their own affiliated projects, concerns were raised about the lack of transparency surrounding how LSEs evaluate interconnection requests and allocate deliverability. The absence of standardized disclosure requirements limits developers' understanding of how allocation decisions are made and undermines confidence in the fairness of the process.

While stakeholder concerns about LSE involvement are valid, CAISO's decision to elevate their role in the IP is arguably the most practical solution given current institutional and structural

constraints. In California, LSEs maintain uniquely close relationships with regulatory agencies and policymakers, which has positioned them to take on a more central role in project selection compared to their counterparts in other regions. This expanded authority reflects the broader policy framework in which LSEs are expected not only to meet procurement mandates but also to help guide the system toward cleaner and more reliable energy sources. While this arrangement is not without tradeoffs, most stakeholders acknowledge that transmission deliverability remains a constrained and costly resource.

Procurement and Study Timelines

Another frequently cited challenge across interviews was the misalignment between procurement timelines and interconnection study requirements. While many factors contribute to the growing congestion of the IQ, a particularly persistent issue was related to how new generation is procured.

Multiple interviewees identified the five-year minimum term requirement for a PPA as a key barrier. This requirement, which determines eligibility for meeting commercial readiness milestones, creates what many stakeholders describe as a classic “chicken-and-egg” dilemma: developers cannot secure a PPA without demonstrating project maturity, but achieving that level of certainty often depends on already having a PPA in place. The result is a bottleneck in project development and increased attrition risk for otherwise viable proposals.

While there was broad agreement that this issue must be addressed, the options for doing so are limited. Procurement processes are primarily governed by the CPUC and shaped by broader state policy goals, leaving CAISO with little formal authority to intervene. Nevertheless, interviewees

emphasized the importance of stronger coordination between CAISO and CPUC. By actively engaging in policy development and providing input on proposed procurement mechanisms through programs like the RCPMP, CAISO can help shape policies that align with and support its interconnection reforms. Better integration between procurement timelines and study requirements may ultimately improve project viability and reduce unnecessary churn in the queue.

Table 13. Interview Summary: Stakeholder Sentiments

Common Themes	Shared Sentiments	Developers	Regulators	LSE
IQ Reform	- Queue reform is necessary and overdue.	- Prefer gradual, predictable rollouts to allow adjustment to financial and planning models.	- Favor immediate and stricter queue filters to rapidly remove speculative and duplicative projects, enhancing overall reliability and planning efficiency.	- Support decisive measures like the LSE points system as effective, practical means to rapidly shrink the queue despite acknowledging the burdens on LSEs.
	- Excessive speculative projects burden system.	- Concern about overly restrictive entry criteria discouraging development.		
	- Incremental improvements helpful.			
Transparency and Data Accessibility	- Improved transparency is universally desirable.	- Primarily interested in transparent deliverability scoring criteria, detailed heat maps, and	- Emphasize structured governance and careful data management, balancing transparency with	- Want more transparent signals from developers about project viability and realistic

Common Themes	Shared Sentiments	Developers	Regulators	LSE
		clearer substation congestion info for competitive planning.	confidentiality and reliability concerns.	timelines, as opposed to purely speculative interest.
	- Accurate, accessible, and detailed data sharing needed for efficient planning.	- Prefer timely, precise data availability.	- Aim to avoid speculative reactions from developers based on preliminary data.	- Seek standardized naming conventions, publicly accessible project contact details, and clearer GIS mapping tools.
Deliverability Allocation Processes	- Current deliverability allocation process perceived as challenging but essential.	- View current deliverability allocation as unclear, unfairly benefiting certain projects (IOU-owned or preferred projects).	- Consider early reservation of deliverability essential for strategically significant public planning (e.g., offshore wind, reliability-critical projects).	- Understand the necessity of current system but express frustration at inefficiencies and high costs of administering the LSE scoring processes.
	- Recognized need for strategic allocation improvements.	- Highly concerned about securing PPAs without clear deliverability status.	- Believe the CPUC's structured allocation approach reduces long-term risk and financial uncertainty.	- Prefer allocations supporting co-located generation plus storage projects due to reliability concerns.

Common Themes	Shared Sentiments	Developers	Regulators	LSE
Procurement & Interconnection Study Timing	- Misalignment between procurement cycles and interconnection studies leads to systemic inefficiencies and delays.	- Cannot afford long waits for procurement commitments, risking financial losses and project feasibility.	- Recognize misalignment, but their planning cycles and mandates (TPP, IRP, RA processes) necessitate careful, structured procurement timelines.	- Expresses frustration at inability to act decisively without CPUC-approved procurement.
	- Alignment needs significant improvement.	- Need clearer procurement signals earlier to minimize speculative investment risk.	- Acknowledge the inherent friction between planning mandates and developer timelines but see limited room for faster adjustment.	- Emphasizes reliance on formalized procurement cycles, delaying decision-making and causing developer frustration.
Strategic Resource Planning	- Strategic resource planning essential for clear long-term reliability and decarbonization goals.	- See busbar mapping and RCPPP as helpful but insufficiently detailed or flexible for rapid market shifts.	- Strong advocates for structured strategic planning (IRP, RCPPP).	- View structured planning tools positively but see ongoing mismatches between theoretical resource planning and actual queued projects.
	- Busbar mapping and strategic procurement tools recognized as beneficial, despite imperfections.	- Desire clearer planning signals from CPUC and greater flexibility in responding to market conditions.	- View busbar mapping as essential tools to direct market outcomes and deliverability allocation strategically.	- Frustrated by modeling assumptions (RESOLVE) and real-world feasibility mismatches (e.g., offshore wind, Wyoming wind).

Common Themes	Shared Sentiments	Developers	Regulators	LSE
			- Recognize need for better alignment between modeling and practical realities.	
Regulatory Stability and Predictability	- Regulatory stability highly desirable for effective long-term planning and investment.	- Strongly emphasize need for predictable, stable regulations; frequent changes cause significant financial risk and uncertainty.	- Acknowledge rapid regulatory changes as somewhat inevitable due to changing policy objectives.	- Express frustration with rapid regulatory shifts complicating procurement timelines and project management.
	- Frequent, rapid regulatory changes seen as disruptive.	- Highly concerned about quickly shifting rules impacting investment viability.	- Aim to mitigate disruption through structured frameworks and careful communication.	- Desire clearer advance notice of regulatory changes to allow for better internal planning and budgeting.
			- Recognize tension between regulatory responsiveness and market stability.	

3.2.1 – Stakeholder Perspective Comparisons

While there is broad consensus among stakeholders that reform is urgently needed, the underlying motivations for that agreement often diverge. Developers and LSEs both advocate for greater transparency, but developers fear losing competitive advantage while LSEs require

information access to meet procurement mandates. Similarly, regulators and LSEs support structured procurement frameworks and planning instruments like busbar mapping, but developers view these tools as inflexible and sometimes misaligned with market conditions. On deliverability, all parties recognize its centrality to project viability, yet they differ in their preferred mechanisms for allocation. Developers seek equitable access, while regulators prioritize strategic planning outcomes. These divergences highlight the importance of reforms that not only streamline procedures but also reconcile the distinct operational and strategic priorities of each stakeholder group. Future iterations of the IP must strike a balance between predictability for developers, flexibility for LSEs, and accountability for regulators to ensure alignment with California's energy goals.

Chapter 4: Recommendations and Conclusion

CAISO, working in coordination with other stakeholders, has taken active steps to reform the IP in response to the challenges observed in recent cluster studies. While these efforts reflect a genuine attempt to streamline operations and prioritize viable projects, many of the most persistent barriers ultimately trace back to a more fundamental constraint: the limited availability of transmission deliverability. Based on interviews, document analysis, and a review of recent policy developments, this chapter offers several recommendations intended to support more effective coordination among the diverse participants in California's interconnection system.

4.1 – Expanded Stakeholder Data Exchange Platforms

As emphasized by many interviewees, improved data accessibility remains a central need across all stakeholder groups. While CAISO, CPUC, and CEC have made strides in coordinating planning, gaps in data availability, clarity, and usability continue to hinder timely decision-making. The current system of stakeholder engagement and open transmission forums provides foundational transparency, but often lack mechanisms for ongoing, project-level exchange between participants.

To address these limitations, regulators could consider establishing a voluntary data-sharing platform where stakeholders can opt in to exchange timely, project-relevant information. For example, SB 1174 requires LSEs to report delayed transmission projects to the CPUC. This information is not broadly disseminated, yet it could be of substantial value to developers seeking to site projects in affected areas. Similarly, CAISO's heat map that was introduced in

response to FERC Order 2023 offers a promising base layer for geospatial decision-making. Its utility could be further enhanced through the addition of public-facing overlays contributed by LSEs, the CEC, and project developers.

Although participation in such a system would likely be voluntary at the outset, interview responses suggest that many stakeholders would welcome a more structured, centralized mechanism for exchanging critical planning information. While implementation would raise questions about data standardization, temporal resolution of data, and commercial sensitivity, the potential benefits that could be seen through increased coordination, earlier identification of risks, and reduced speculative activity are substantial and not to be underestimated.

4.2 – Increased Intra-Agency Integration of Regulatory Planning Initiatives

Looking at the most recent IPE changes, there is evidence that the regulatory organizations of the IP are taking steps to further centralize an already centralized process. The ambitious energy goals of the state have manifested in an electrical grid that is reliant on a continuously increasing level of planning and regulation, and it has become more and more clear through the direction of policy decisions being made that this will remain the case for the near future. The CPUC and CAISOs commitment to reserving deliverability for specific resource types, the newly proposed RCPMP program, the deepening involvement of LSEs in the cluster study process, the tightening requirements for resources to qualify as RA or RPS generation and the corresponding increase in the importance of these factors for acquiring deliverability allocations are all indications that future policy will be crafted to support these wider energy goals,

As this integration deepens, it is critical that coordination between agencies becomes more formalized. While current collaboration efforts are meaningful, the absence of binding frameworks for inter-agency communication can lead to misalignments in procurement targets, transmission build-out expectations, and scoring criteria. Future iterations of the IRP, TPP, and IPE would benefit from more prescriptive protocols that ensure consistency across these planning cycles. This includes setting expectations for data exchange, aligning modeling assumptions, and clarifying how deliverability priorities are reconciled with local permitting and siting constraints.

4.3 – Expanded Avenues for TPD Bidding

Among developer-submitted comments during the IPE stakeholder process, one of the more frequently proposed reforms was the introduction of a deliverability exchange, an open platform where stakeholders could bid on or transfer TPD allocations. While a form of auction already exists to break ties during study qualification, its application is narrow and reactive. Broadening this mechanism could help alleviate a commonly observed inconsistency; deliverability rights often remain tied to projects that become delayed, downscaled, or canceled, leaving valuable capacity underutilized.

An expanded deliverability exchange that operates on a recurring basis and is open to both sellers and buyers could enhance resource allocation, promote project flexibility, and improve the overall efficiency of infrastructure use. In such a system, developers approaching commercial operation but lacking deliverability could actively bid on available allocations, while those holding unneeded capacity could recoup value by releasing it to others. Transparent clearing

prices would provide insights into substation-level scarcity, guiding both investment and planning decisions more effectively than static allocations alone.

A deliverability exchange would not have to launch immediately, or at full scale. To mitigate risks and ease stakeholder concerns, CAISO could pilot this mechanism in a limited zone before considering a broader rollout. Such a pilot could test auction design, settlement logistics, and participant behavior, while allowing regulators to calibrate parameters such as bid limits, eligibility criteria, and participation costs. If executed carefully, this reform could represent a pragmatic evolution in TPD policy that better aligns infrastructure use with market readiness and development timelines.

4.4 – Conclusion

The IP and IQ has become a barometer of sorts to California's wider energy transition, with the projects populating it acting as indicators for commercial interest, policy direction, and infrastructure adequacy. This report offers a snapshot of a system in evolution, one where well-intentioned reforms are underway but grid congestion, study delays, and procurement-process mismatches continue to inhibit progress.

The study of current IQ data showed that the time a typical project remains under study has lengthened noticeably since the last large tariff overhaul. Part of that growth reflects the sheer volume of renewable and storage capacity seeking to connect, but another less obvious part comes from misaligned process milestones. Procurement decisions and other commercial interest flags are not always available, and when developers must decide whether to enter a cluster window many end up filing first and hope to contract later. At the same time, local permits and

federal land reviews follow their own publishing cycles, meaning even projects that clear the ISO's technical hurdles can be held back by delays unrelated to grid operation. The result is a queue that looks crowded on paper but delivers new megawatts to the grid more slowly than planners anticipate.

Interviews and workshop notes added a human dimension to those timelines. Developers shared that incomplete visibility into real-time study progress forces them to rely on second-hand updates or compile their own databases, which raises costs and favors established firms. Some stakeholders reported that they struggle to time solicitations, sometimes committing to projects that later withdraw or trying to buy capacity only to find the queue fully inundated. Regulators pointed out that the lack of a single source for transmission upgrade status leaves counties guessing when or if a project in their area will move ahead. Across these conversations, two sentiments were frequently observed: stakeholders are willing to share information, and many believe that there exist modest procedural fixes that could potentially add optimization without the need for large-scale and disruptive changes.

The policy evaluation portion of the report assessed recent ISO reforms and found clear action being taken but also found that many of these measures will require refinement. For example, the heat map highlights thermal constraints yet omits permitting risk, fuel supply context, or local land-use overlays that matter just as much when assessing an area for potential development. Several potential fixes are already in the works as well. Proposals for programs like the CPUC's RCPPP would allow market signals to reach developers before speculative filings swell the queue. Future work can build on this foundation with richer and more consistent datasets. CAISO's study database already logs each milestone; cleaning and publishing those timestamps would allow neutral observers to track cycle times with precision. The CPUC's contract tracker,

if released in aggregated form, could show how offtake certainty affects study outcomes. County permitting portals increasingly publish review timelines online; standardizing those feeds would let planners anticipate non-electrical delays. Federal geospatial layers from land ownership to species habitat can be merged with queue locations to flag siting risks early.

While discussing policy change however, it is important to keep in mind that excessive changes risk unnerving the developers needed to build out our future clean energy capacity. Developers raise capital on the expectation that study timelines, cost liabilities, and deliverability rules will remain broadly predictable; when those parameters change mid-cycle, lenders reconsider risk, insurance costs rise, and smaller firms can lose access to credit altogether. A queue reform that accelerates studies but leaves upgrade cost recovery unclear, or a new scoring rubric that reshuffles project priorities may push away investors until things return to a more stable state.

This runs the additional risk of slowing the rate of development when it matters most. The solution will lie in balancing decisive action with clear transition pathways that facilitate the job of developers. Grandfathering clauses that honor prior commitments, buffer periods that give stakeholders time to adjust to changes, transparent data releases that show how new measures perform, and other such actions will be necessary to maintain the goodwill of all stakeholders. Striking that balance protects ratepayers from delay while assuring developers and their backers that the ground will not shift beneath projects already in motion.

In sum, the IQ is not merely a lineup of projects waiting for studies; it is a real time look into the system that touches nearly every aspect of grid planning and clean-energy development. The findings here suggest that practical, collaborative action beats sweeping overhaul and that informed transparency remains the strongest lever for unlocking the queue's latent potential. With time, effort, and patience, it is likely that we will be able to overcome these obstacles and

ensure that our energy grid remains robust and agile enough to not just survive, but thrive in the volatile energy landscape we find ourselves in today.

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Appendices

Appendix A — Generalized Interview Questions

The following questions were provided to stakeholders before, and answered during a series of semi-structured interviews conducted for this study. The interviews were informal in structure, with flexibility to adapt follow-up questions based on the stakeholder's expertise and experience.

A.1 Stakeholder Role and Background

1. Can you describe your role in your organization and how it relates to interconnection and/or transmission planning?
2. What has been your experience working within or alongside the interconnection process in California?
3. Have you worked on projects spanning both retail and wholesale interconnection contexts?

A.2 Interconnection Queue and Policy Reform

4. How has your organization experienced the recent interconnection queue congestion, particularly in Clusters 14 through 16?
5. What is your view on the effectiveness of the CAISO Interconnection Process Enhancements (IPE), especially the Track 2 and Track 3 reforms?
6. Do you believe the changes made to the cluster study process, such as scoring thresholds, study caps, or prioritization criteria have been effective in filtering speculative or non-viable projects?

A.3 Deliverability and Developer Risk

7. How do concerns over transmission deliverability affect your organization's project development or investment strategy?
8. What is your opinion on the current deliverability scoring system, especially the emphasis on LSE support, site control, and PPA status?
9. How does the uncertainty surrounding deliverability allocations and capacity payments impact project financing and timelines?

A.4 Institutional Coordination and Planning

10. How would you characterize the relationship between the CPUC and CAISO in coordinating transmission planning, interconnection, and procurement?
11. To what extent does CPUC's busbar mapping and portfolio modeling influence CAISO's Transmission Planning Process (TPP)?
12. Do you believe the current level of coordination between CPUC and CAISO is sufficient to meet the state's transmission and renewable energy goals?

A.5 Interconnection Barriers and Structural Challenges

13. From your perspective, what are the most significant barriers or bottlenecks in the current interconnection process?
14. Are there specific regulatory, financial, or geographic constraints that have made development more difficult in recent years?

15. What changes or improvements would most help streamline or de-risk the interconnection process?

A.6 Procurement and Resource Adequacy

16. How do CPUC procurement orders (e.g., the RCPMP) or RA market structures influence your interconnection and siting decisions?
17. Have power purchase agreement (PPA) requirements presented challenges, particularly in relation to IPE scoring or project viability?
18. Do you believe the current procurement timeline is well-aligned with the interconnection study and queue management process?

A.7 Regional Disparities and Locational Constraints

19. Are there specific regions in California (e.g., Humboldt, Southern California merchant zones) where interconnection or transmission constraints are especially challenging?
20. How does geographic variation in load, infrastructure, and policy influence your organization's development strategy?

A.8 Transparency, Access, and Data Availability

21. In your experience, is there sufficient public access to interconnection queue information, transmission maps, and contact details for developers?
22. Would greater transparency such as clearer project metadata, developer identification, or spatial visualization tools help improve stakeholder engagement or reduce duplicative efforts?

A.9 Broader Reflections and Recommendations

23. Do you believe the current trajectory of CAISO and CPUC coordination and reforms is adequate to meet California’s long-term energy and climate targets?
24. What policy or process changes would most improve the overall effectiveness and equity of the interconnection system?
25. Is there anything else you believe is important for researchers or policymakers to understand about the current state of California’s interconnection process?

Appendix B: IPE 2021 Shareholder Comment Format

The following tables summarize the structured comment format used in the 2021 Interconnection Process Enhancements (IPE) initiative conducted by the California ISO. Stakeholders were invited to submit feedback through two main tracks (Track 1 and Track 2), with specific prompts tied to individual proposal sections.

Track 1 – Near-Term Enhancements

Stakeholders were asked to respond to the following prompts:

1. Please provide a summary of your organization’s comments on the Interconnection Process Enhancements (IPE) draft final proposal for Phase 1: Near-Term Enhancements.
2. Provide your organization’s comments on the ISO’s proposal to remove the downsizing window and simplify downsizing request requirements (Section 3.1).

3. Provide your organization's comments on the ISO's proposal for revising the Transmission Plan Deliverability (TPD) Allocation process (Section 3.2).
4. Provide your organization's comments on the ISO's proposal for developing an emergency generation interconnection process (Section 3.3).
5. Provide your organization's comments on the ISO's proposal for determining if site exclusivity should be required to progress into the Phase II study process (Section 4.1).
6. Provide your organization's comments on the ISO's proposal for the expanded errors and omissions process regarding network upgrade changes after Financial Security postings (Section 5.1).
7. Provide your organization's comments on the ISO's proposal for clarifying the definition of Reliability Network Upgrade (Section 5.2).
8. Provide your organization's comments on the ISO's proposal for transferring Participating Transmission Owner (TO) Wholesale Distribution Access Tariff (WDAT) projects into the ISO queue (Section 5.3).
9. Provide your organization's comments on the ISO's proposal for allowing changes to sites and Points of Interconnection (POIs) during Interconnection Request (IR) validation (Section 5.4).
10. Provide your organization's comments on whether parked projects should be allowed to submit any type of Material Modification Applications (MMAs) while parked (Section 5.5).

11. Provide your organization's comments on the ISO's proposal to add due dates for curing deficiencies in Appendix B to avoid delays in starting Phase II studies (Section 6.1).
 12. Provide your organization's comments on the ISO's proposal for modifications to commercial viability criteria (Section 6.2).
 13. Provide your organization's comments on the ISO's proposal for expanding deliverability transfer opportunities (Section 6.3).
 14. Provide your organization's comments on the ISO's proposal to require documentation from third-party gen-tie owners as part of the IR (Section 6.4).
 15. Provide your organization's comments on the ISO's recommendation that RIMS be used consistently for all post-validation project documentation (Section 6.5).
 16. Please provide additional comments on the IPE Final Proposal – Phase 1 not addressed above.
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Track 2 – Structural and Policy Reforms

Stakeholders were asked to respond to the following prompts:

1. Please provide a summary of your organization's comments on the IPE 2021 Phase 2 Final Proposal.
2. Provide your organization's comments on Section 3.1 – Transparency enhancements.
3. Provide your organization's comments on Section 3.2 – Revisiting the criteria for PPAs to be eligible for a TPD allocation.

4. Provide your organization's comments on Section 5.1 – Whether the ISO should reconsider alternative cost allocation for upgrades to local (sub-200 kV) systems benefiting other-than-local customers.
5. Provide your organization's comments on Section 5.2 – ISO as an Affected System: how base cases are determined and upgrade costs are allocated.
6. Provide your organization's comments on Section 5.2 – Whether the ISO should have the ability to terminate the GIA earlier than seven years if a project is not progressing toward permitting and construction.
7. Provide your organization's comments on Section 6.1 – Process for when a developer issues a notice to proceed to the PTO, including planning for all upgrades required for Full Capacity Deliverability Status (FCDS), even those triggered by multiple projects.

Appendix C: IPE 2023 Shareholder Comment Format

The following outlines the stakeholder comment format for the 2023 Interconnection Process Enhancements (IPE) initiative, as facilitated by the California ISO. Stakeholder input was solicited in three tracks: (1) Tariff review, (2) procedural and policy feedback, and (3) structural modifications to prioritization and deliverability systems.

Track 1 – Draft Tariff Language Review

Stakeholders were asked to respond to the following prompts:

1. Please provide a summary of your organization’s comments on the Interconnection Process Enhancements 2021 – Phase 1 draft tariff language.
 2. Please upload redlined tariff language using the “attachments” field.
-

Track 2 – Procedural Feedback and Initiative Planning

Stakeholders were asked to respond to the following prompts:

1. Please provide your organization’s comments on the Interconnection Process Enhancements (IPE) 2023 Track 2 Final Proposal.
 2. Please specify which section of the proposal your comments address.
 3. Provide your organization’s feedback on the proposed timeline for Track 3 of the IPE 2023 initiative, which focuses on the Transmission Plan Deliverability Allocation process.
 4. The ISO would like your feedback on extending the initiative timeline by targeting the October 3, 2024 ISO Board of Governors meeting for decision.
 5. The ISO is seeking stakeholder feedback on the effectiveness of its recently adopted working group process and its overall contributions to the IPE 2023 proposal development process.
 - Stakeholders were invited to complete a survey at:
<https://www.surveymonkey.com/r/YHWNWC2>
-

Track 3 – Structural Reform and Prioritization Policy

Stakeholders were asked to respond to the following prompts:

1. Please state your organization’s position and provide questions or comments on the intra-cluster prioritization of use of existing System Capacity Deliverability (SCD) and Reliability Network Upgrade (RNU) headroom.
2. Please state your organization’s position and provide your questions or comments on the proposed modifications to the Transmission Plan Deliverability (TPD) Allocation Process.
3. Please state your organization’s position and provide questions or comments on the adjusted second Interconnection Financial Security (IFS) posting requirements for Cluster 14 parked projects.
4. Please state your organization’s position and provide questions or comments on the proposed process for reserving TPD for long lead-time generation and storage resources.
5. Please state your organization’s position and provide any additional feedback on the broader Track 3 initiative, including suggestions for the scope of the next IPE initiative.